Origins and Evolution of Language
Class 3: More iterated learning

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Language change

Language change (as attested in the historical record / inferable from synchronic data) is a consequence of:

- Speakers trying to convey meaning efficiently
- Hearers trying to infer speaker meaning
- Language learners (and everyone else) seeking regularities in the linguistic data they encounter

These processes are inherent to the transmission of language via learning and (ostensive-inferential) use

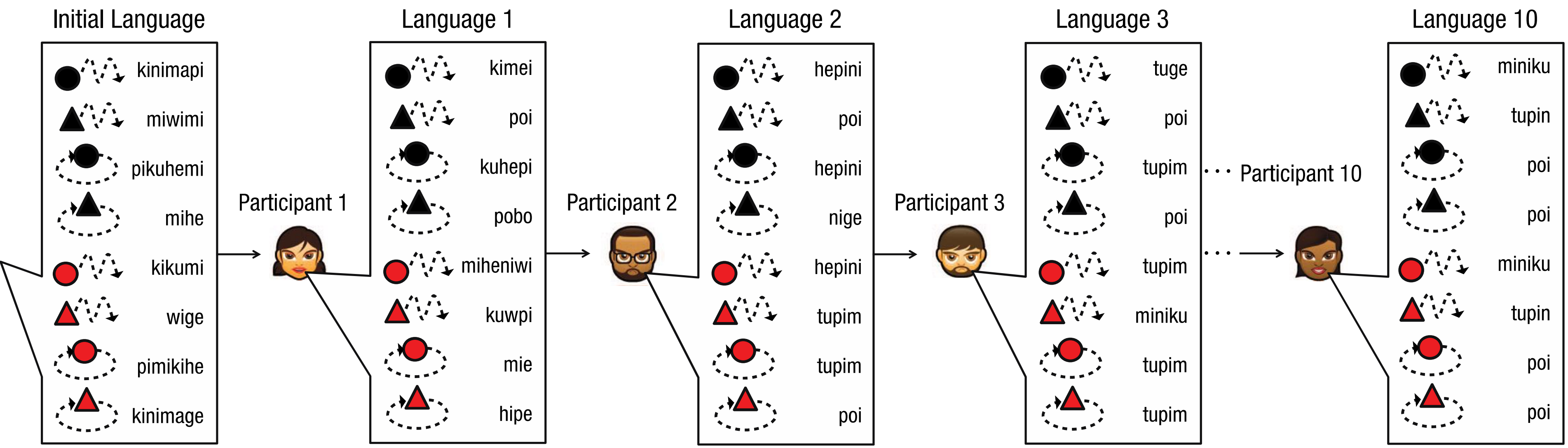
To what extent can these same processes explain the origins of fundamental properties of linguistic systems?

Symbols

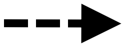
Compositionality

Duality of patterning

Iterated Learning



Final language from chain 1 (!)

	nepa	nepa	nepa	□
	nepa	nepa	nepa	○
	nepa	nepa	nepa	△
	nepa	nepa	nepa	□
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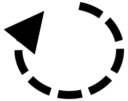
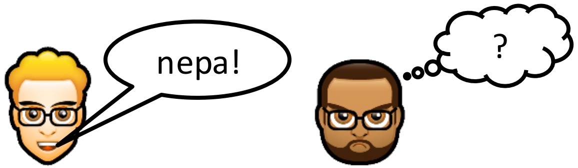
	nepa	nepa	nepa	○
	nepa	nepa	nepa	△

The languages become **degenerate**



Learnability and degeneracy

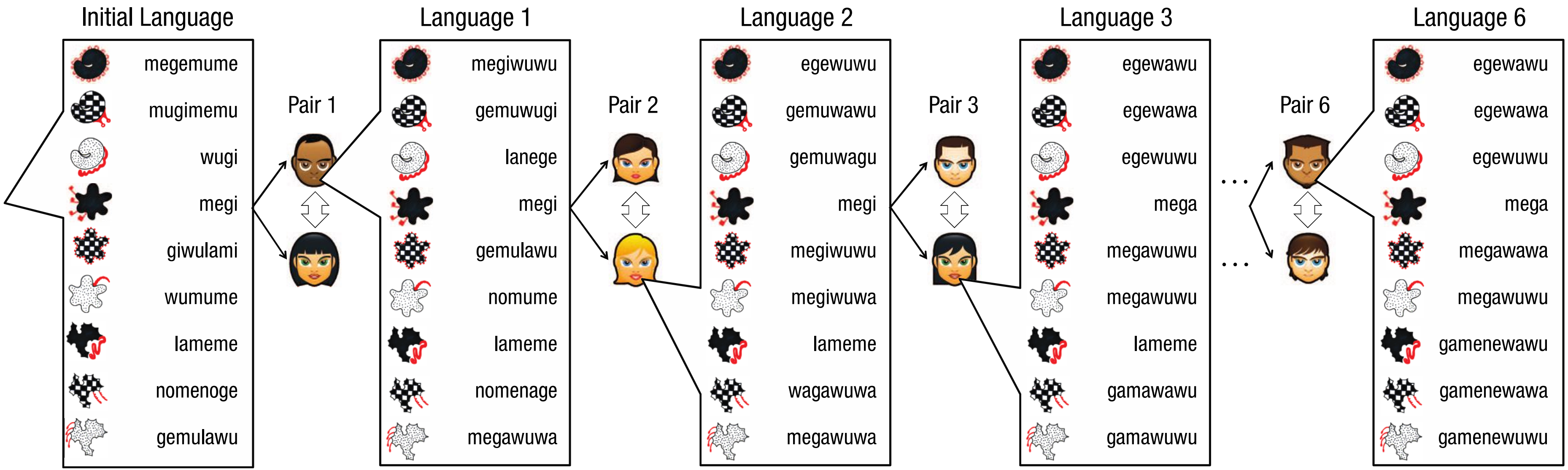
Learners prefer simpler languages

The only pressure in Kirby, Cornish & Smith (2008) Experiment 1 is **learnability**

- The languages don’t need to be **expressive**
- They get very simple

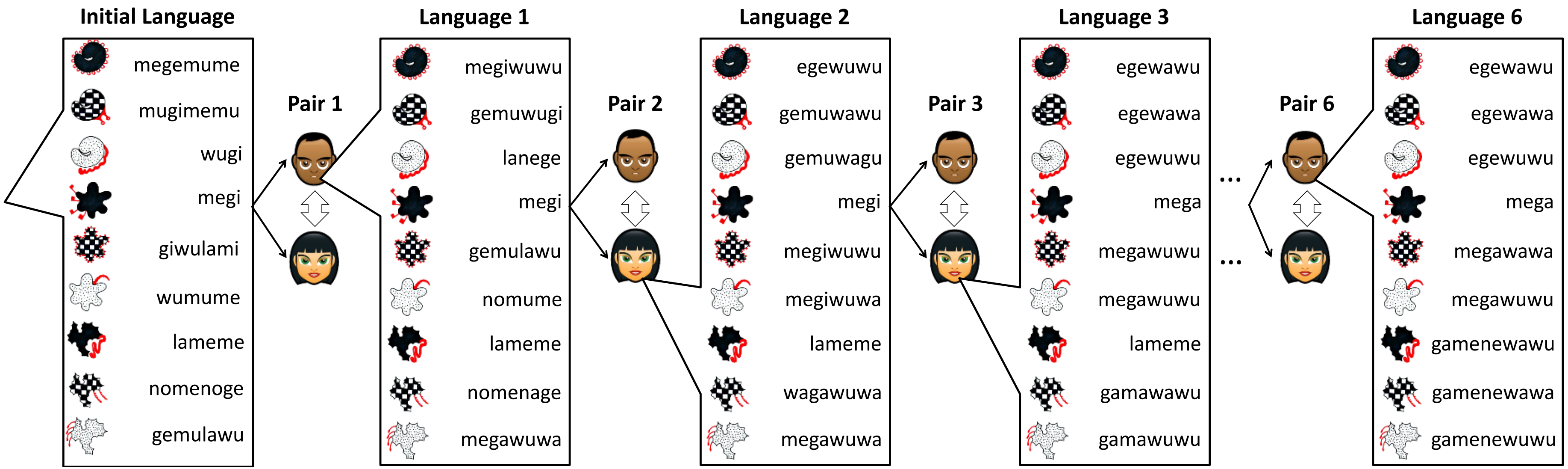
Can we add in a pressure for expressivity?

Kirby, Tamariz, Cornish & Smith (2015): Adding communication



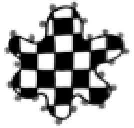
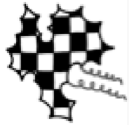
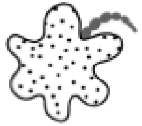
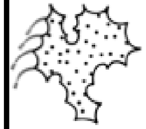
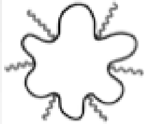
Kirby, S., Tamariz, M., Cornish, H., & Smith, K. (2015). Compression and Communication in the Cultural Evolution of Linguistic Structure. *Cognition, 141*, 87-102.

Kirby, Tamariz, Cornish & Smith (2015): Adding communication, removing learning

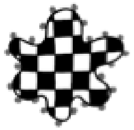
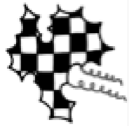
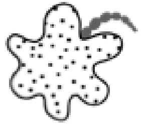
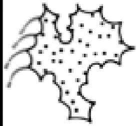
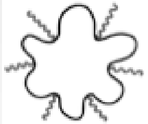


Kirby, S., Tamariz, M., Cornish, H., & Smith, K. (2015). Compression and Communication in the Cultural Evolution of Linguistic Structure. *Cognition, 141*, 87-102.

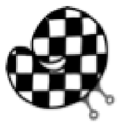
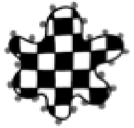
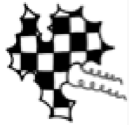
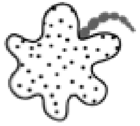
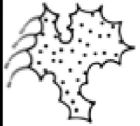
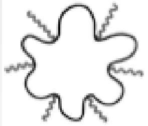
An initial non-compositional mapping

 megemume	 megi	 lameme
 mugimemu	 giwulami	 nomenoge
 wugi	 wumume	 gemulawu
 lamege	 wulamugi	 megiwuwa

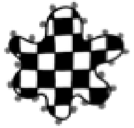
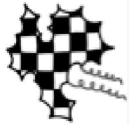
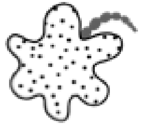
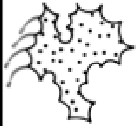
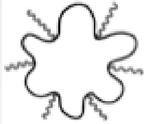
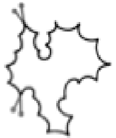
Chain, Generation 1

 megiwuwu	 megi	 lameme
 gemuwugi	 gemulawu	 nomenage
 lanege	 nomume	 megawuwa
 nomege	 wulagi	 wagawu

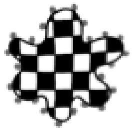
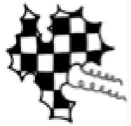
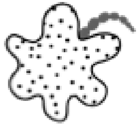
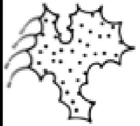
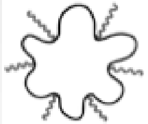
Chain, Generation 2

 egewuwu	 megi	 lameme
 gemuwawu	 megiwuwu	 wagawuwa
 gemuwagu	 megiwuwa	 megawuwa
 ege	 wulagi	 wagawuwa

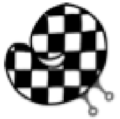
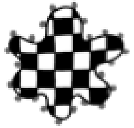
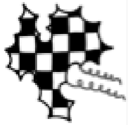
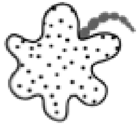
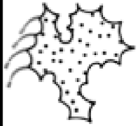
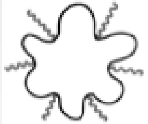
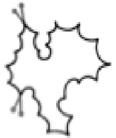
Chain, Generation 3

 egewawu	 mega	 lameme
 egewawa	 megawuwu	 gamawawu
 egewuwu	 megawuwu	 gamawuwu
 ege	 wulagi	 gamene

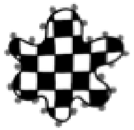
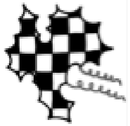
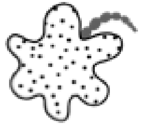
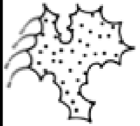
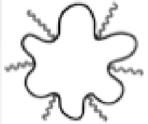
Chain, Generation 4

 egewawu	 mega	 gamenewawu
 egewawa	 megawawa	 gamenewawa
 egewuwu	 megawuwu	 gamenewuwu
 ege	 wulagi	 gamane

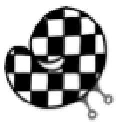
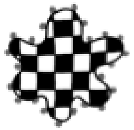
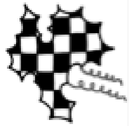
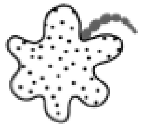
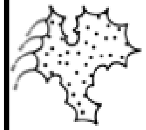
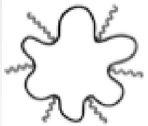
Chain, Generation 5

 egewawu	 mega	 gamenewawu
 egewawa	 megawawa	 gamenewawa
 egewuwu	 megawuwu	 gamenewuwu
 ege	 wulagi	 gamane

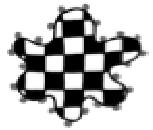
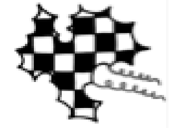
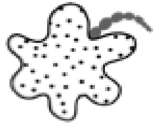
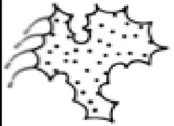
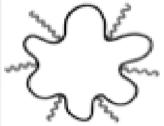
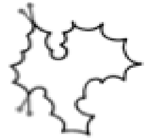
Chain, Generation 6

	egewawu		mega		gamenewawu
	egewawa		megawawa		gamenewawa
	egewuwu		megawuwu		gamenewuwu
	ege		wulagi		gamane

Chain, Generation 6

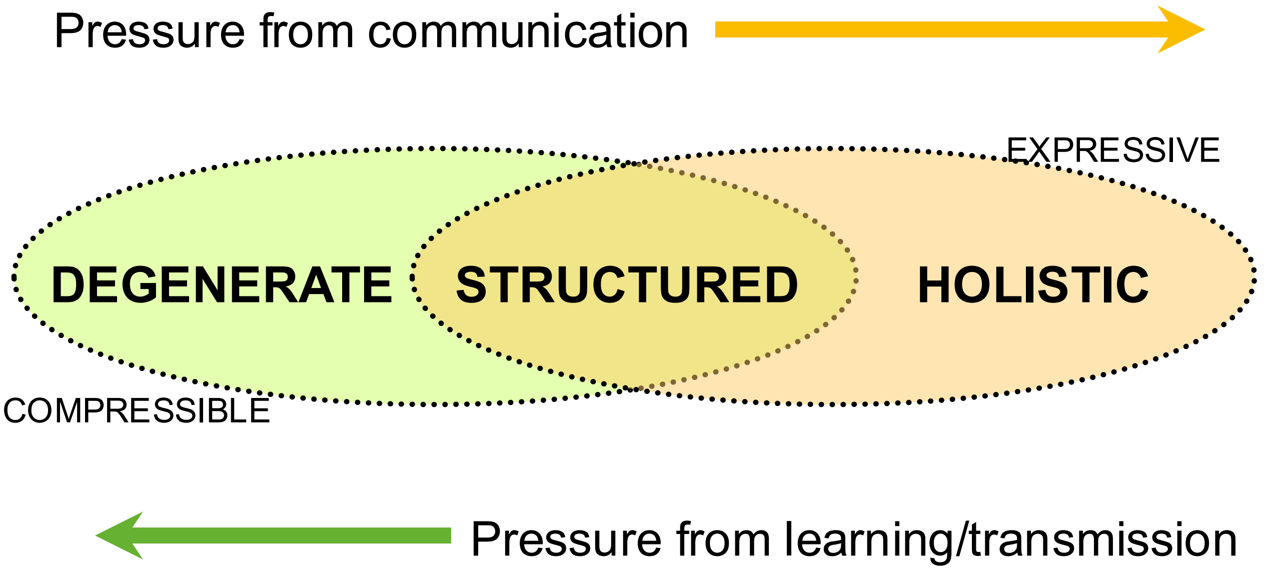
 egewawu	 mega	 gamenewawu
 egewawa	 megawawa	 gamenewawa
 egewuwu	 megawuwu	 gamenewuwu
 ege	 wulagi	 gamane

A final non-compositional language from a dyad

	manunumoko		moko		konu
	wekihumanunu		mokowekihu		lawa
	makihu		mahiku		wekihulawa
	manunumonu		nomu		wekihu

Learnability + expressivity = **structure**

Structure as a trade-off between compressibility and expressivity that plays out over cultural transmission



Questions from the “ask me anything” box

“In Kirby, Cornish & Smith (2008), you demonstrate that lexical items become more structured through evolution (that is, strings becoming more systematic, regular, and even compositional), hence making language more learnable and generalizable. Do you think syntactic structures in natural language also emerged under similar pressures?”

Saldana, C., Kirby, S., Truswell, R., & Smith, K. (2019). Compositional hierarchical structure evolves through cultural transmission: an experimental study. *Journal of Language Evolution, 4,* 83-107.

Questions from the “ask me anything” box

“According to Kirby, Cornish & Smith (2008), do learners from different countries and regions exhibit differences in language structure through unintentionally cultural transmission?”

“Has this been done with child learners?”

Raviv, L., & Arnon, I. (2018). Systematicity, but not compositionality: Examining the emergence of linguistic structure in children and adults using iterated learning. *Cognition, 181*, 160–173.

Questions from the “ask me anything” box

“would these results take away from Chomsky's theory of Language Acquisition Device (LAD) and universal grammar, as it demonstrates that adaptive cultural processes can also explain language acquisition?”

Questions from the “ask me anything” box

“Does the notion of iterated learning presuppose a certain view or theory (for example, nativism vs. constructivism), or is it a theory-neutral term?”

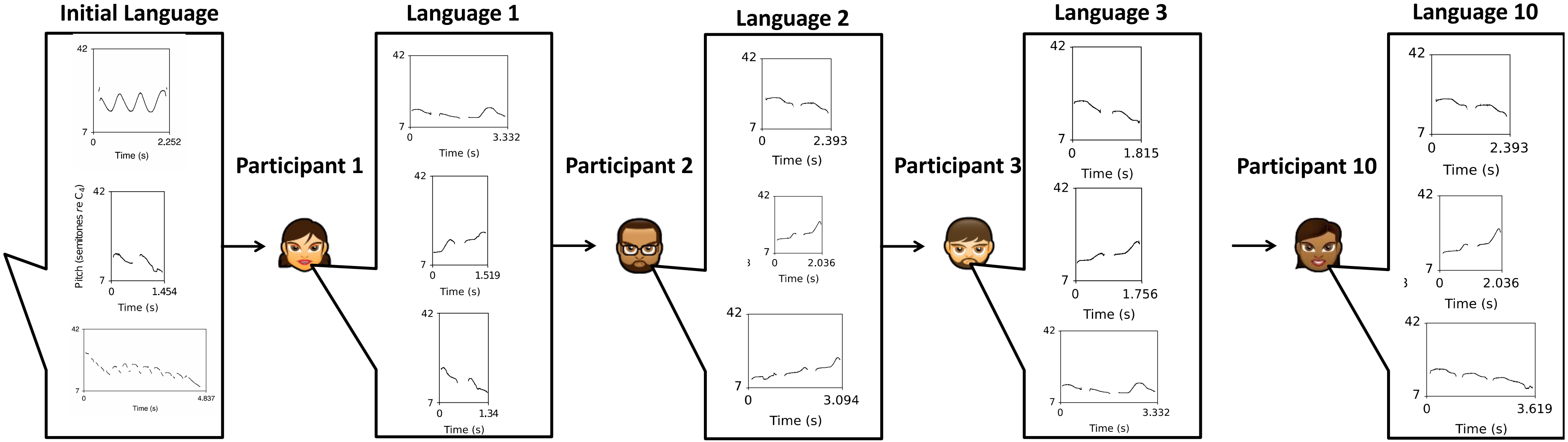
Questions from the “ask me anything” box

“Kirby et al. (2008) argue that culture evolution can deliver the appearance of language without the existence of the designer. I remembered that the invention of Hangul by King Sejong and Qin Dynasty’s “Shu Tong Wen (书同文)” were examples of government initiated top-down reform(maybe it is mixed)? I was wondering what is the nature of linguistic evolution: are top-down and cultural transmission distinct without interactions? ”

Language's communicative power comes (in part) from its **double layer of combinatorial structure**

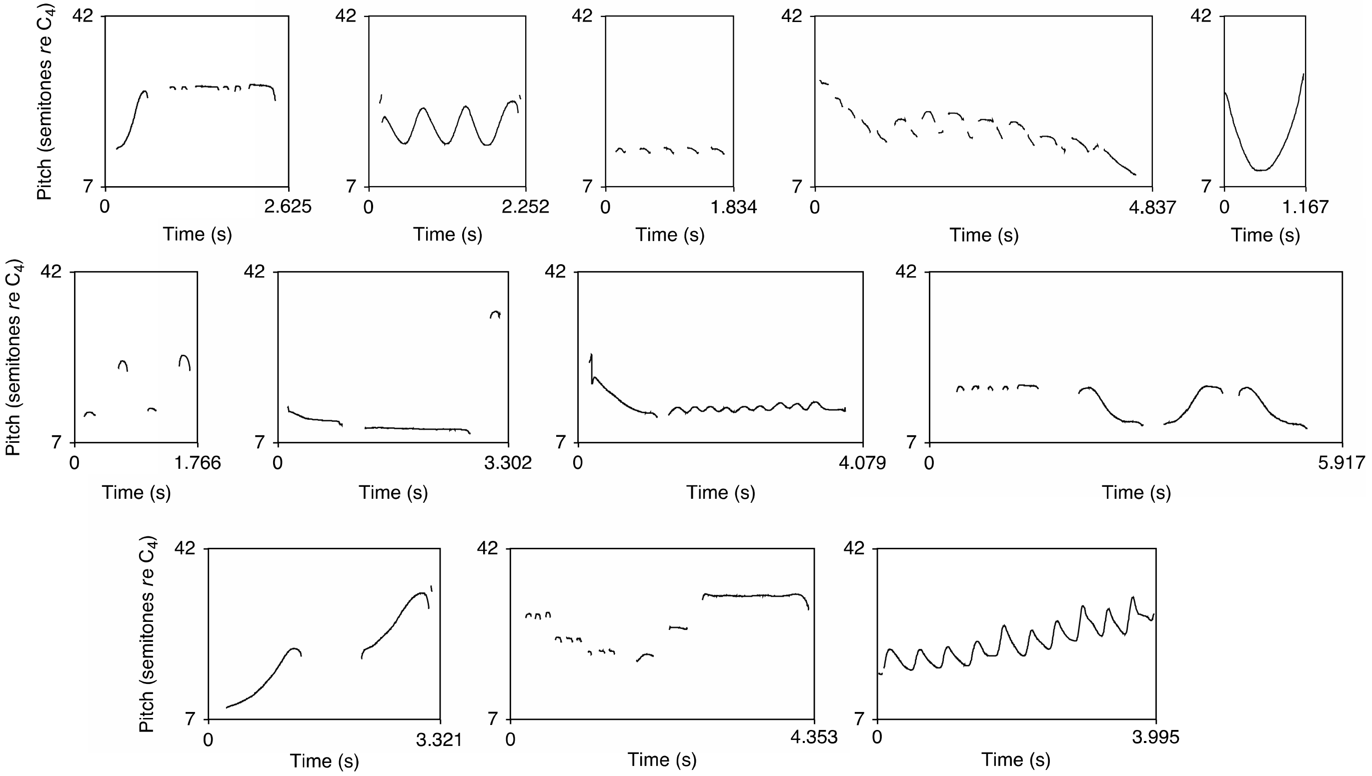
Inventory of meaningless units (10s)	p t d s ð k g ɔ ə a …
Inventory of meaningful units (1000s)	ə *(a)* ðə *(the)* -əd *(past tense)* dɔg *(dog)* kat *(cat)* ðat *(that)* spɔt *(spot)* …
Inventory of meaningful sentences (∞)	*the cat spotted the dog* *a dog spotted the cat* *a cat spotted the dog* *the dog spotted the cat* *the cat spotted the cat that spotted a dog* *the dog spotted the cat that spotted the dog* …

Iterated Learning with whistles

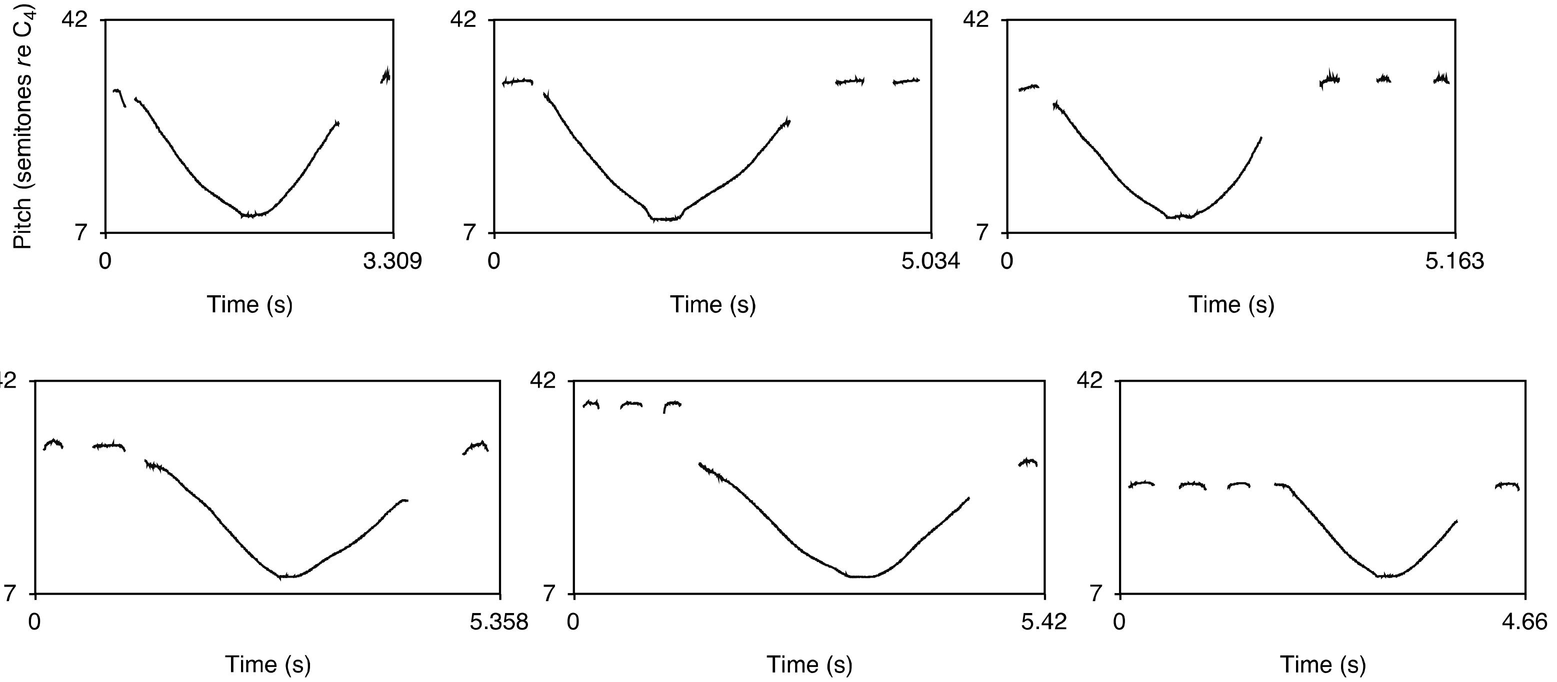


Verhoef, T., Kirby, S., & de Boer, B. (2014). Emergence of combinatorial structure and economy through iterated learning with continuous acoustic signals. *Journal of Phonetics, 43,*

The initial whistle set

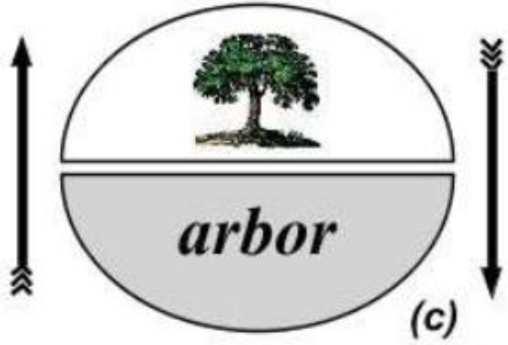


(Part of) A generation 10 whistle set



Where do symbols come from?

- **Icon:** signals resemble meanings
- **Symbol:** *arbitrary* relationship between signal and meaning



Ritualization in the lab, with humans

Repeated interaction in a Pictionary-like communication task

Garrod, S. et al. (2007). Foundations of Representation: Where Might Graphical Symbol Systems Come From? *Cognitive Science, 31*, 961-987

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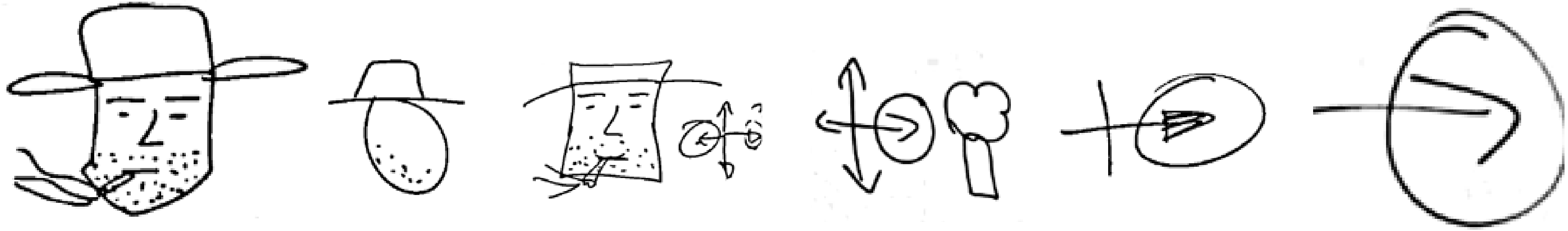
Ritualization in the lab



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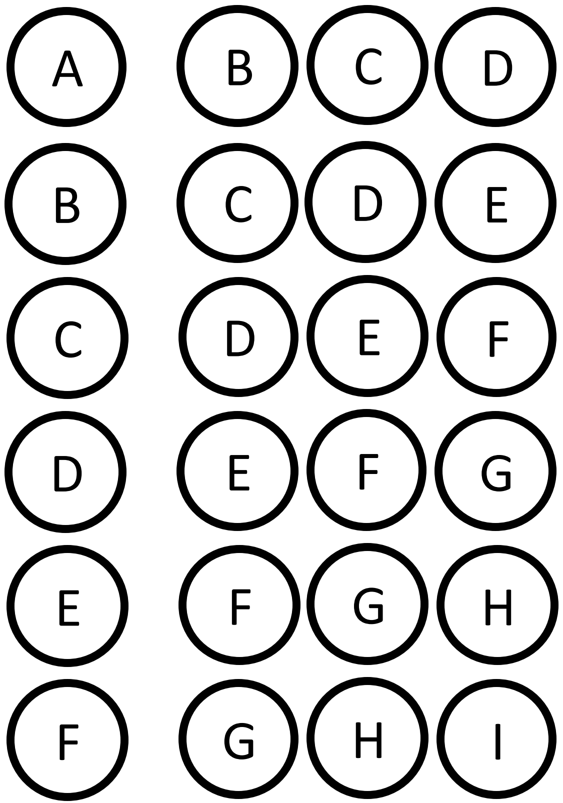
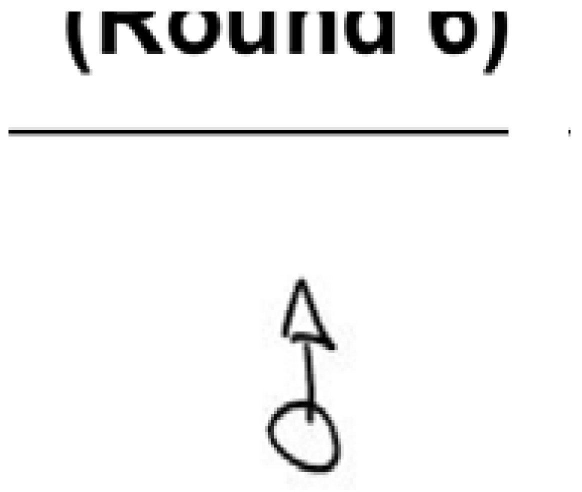
Transmission in laboratory ‘societies’



Caldwell, C. A., & Smith, K. (2012). Cultural evolution and the perpetuation of arbitrary communicative conventions in experimental microsocieties. *PLoS ONE, 7,* e43807.

Transmission in laboratory ‘societies’



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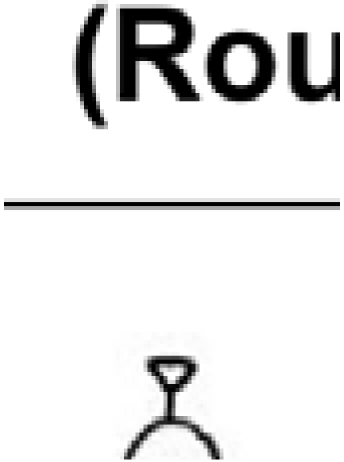

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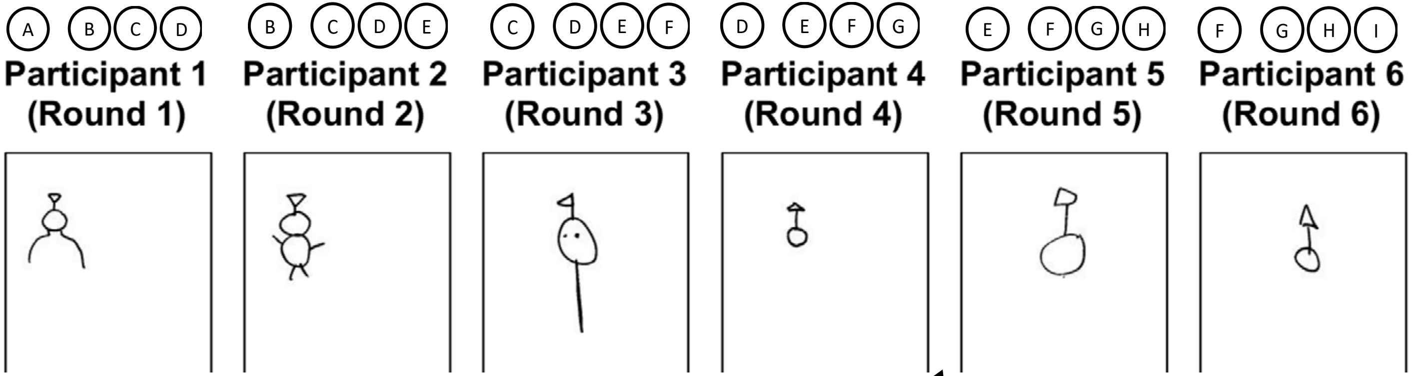


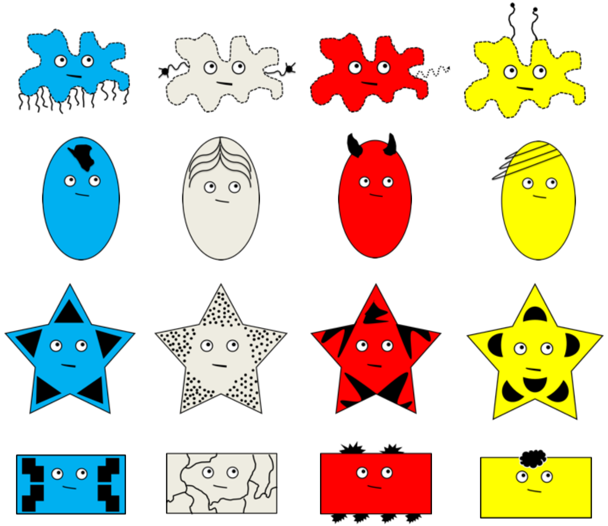
(potentially) truly arbitrary signals

Caldwell, C. A., & Smith, K. (2012). Cultural evolution and the perpetuation of arbitrary communicative conventions in experimental microsocieties. *PLoS ONE, 7,* e43807.

Adaptation to differing communicative constraints

The structure of the communicative task affects the kinds of structures that emerge

E.g. Winters, J., Kirby, S., & Smith, K. (2018). Contextual predictability shapes signal autonomy. *Cognition, 176,* 15-30.



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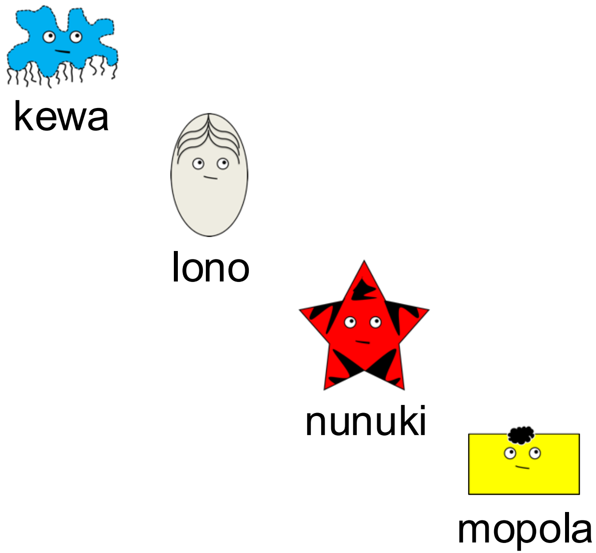

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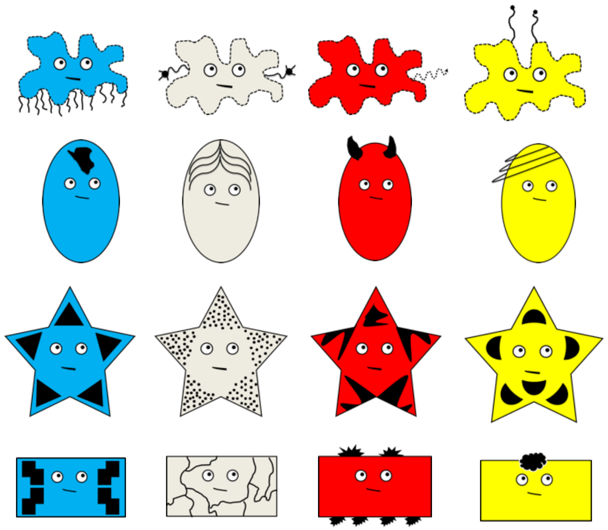
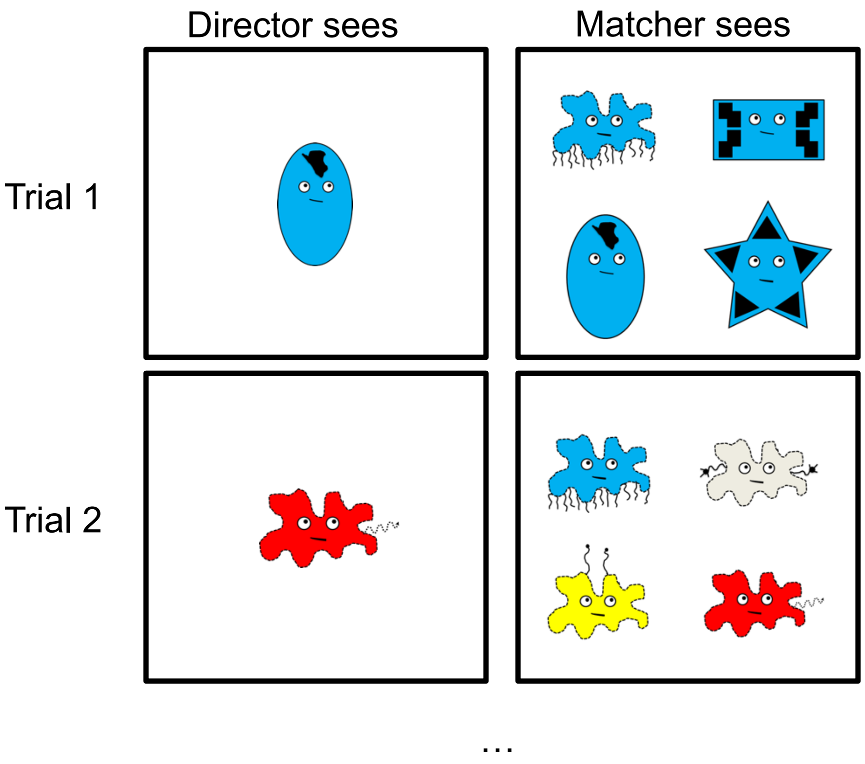
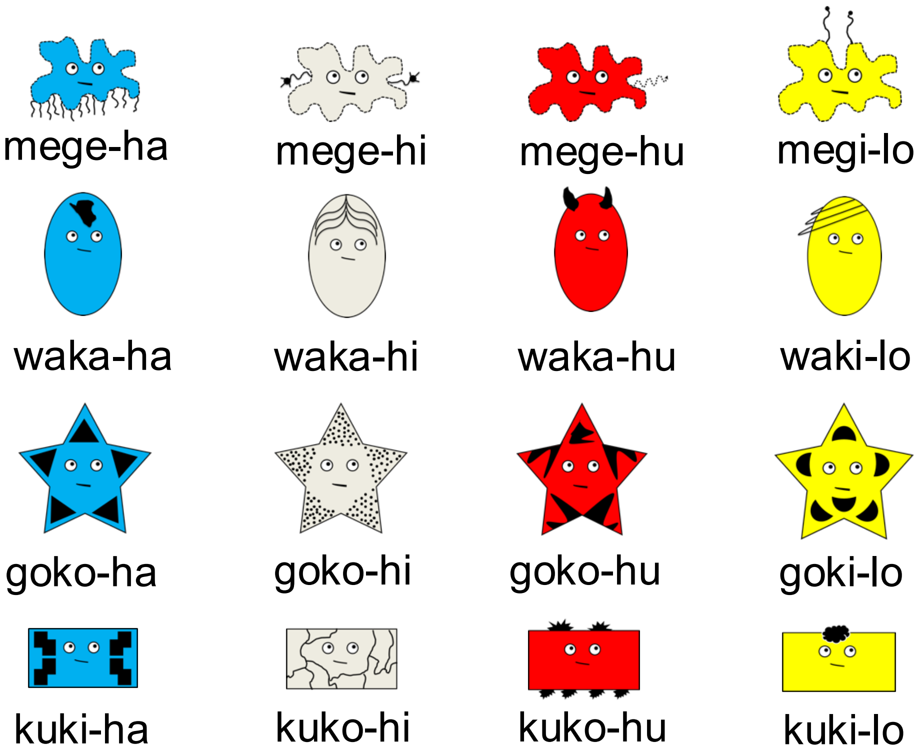
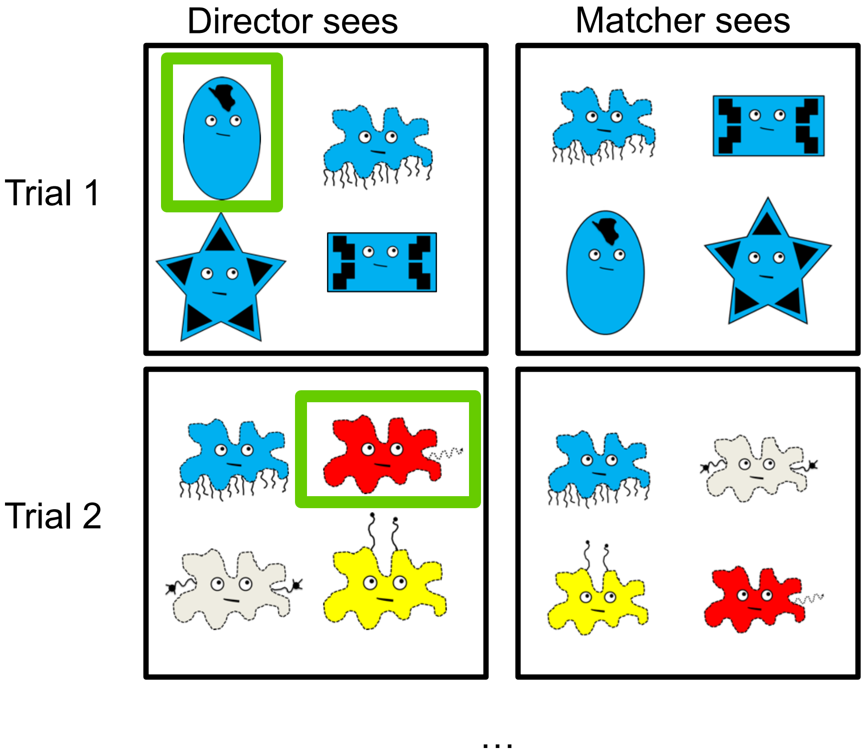
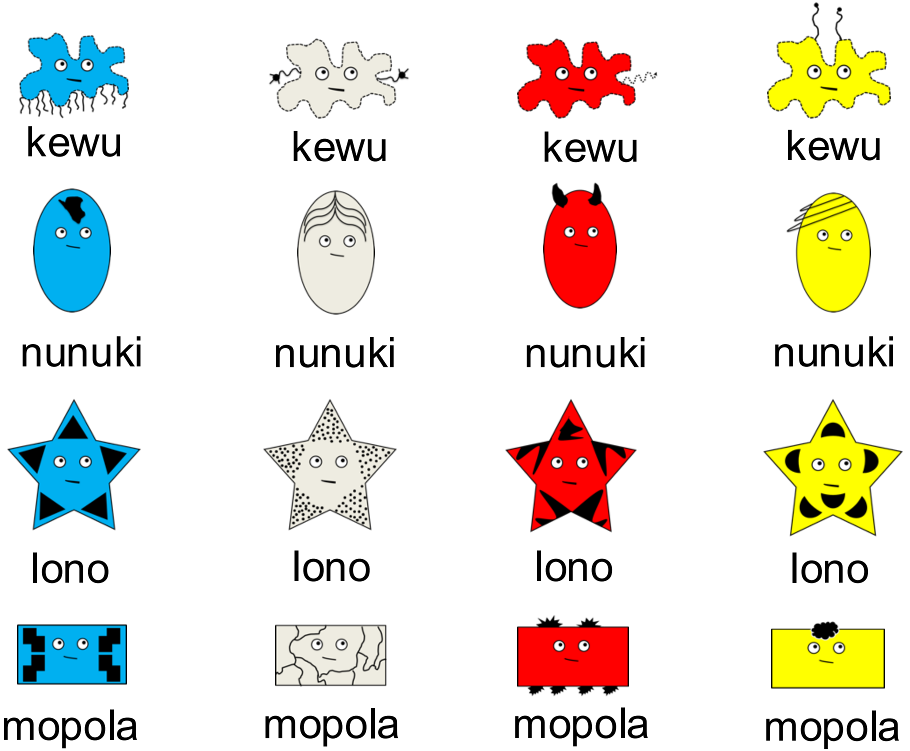



Director sees
Matcher sees
Trial 1
Trial 2
...

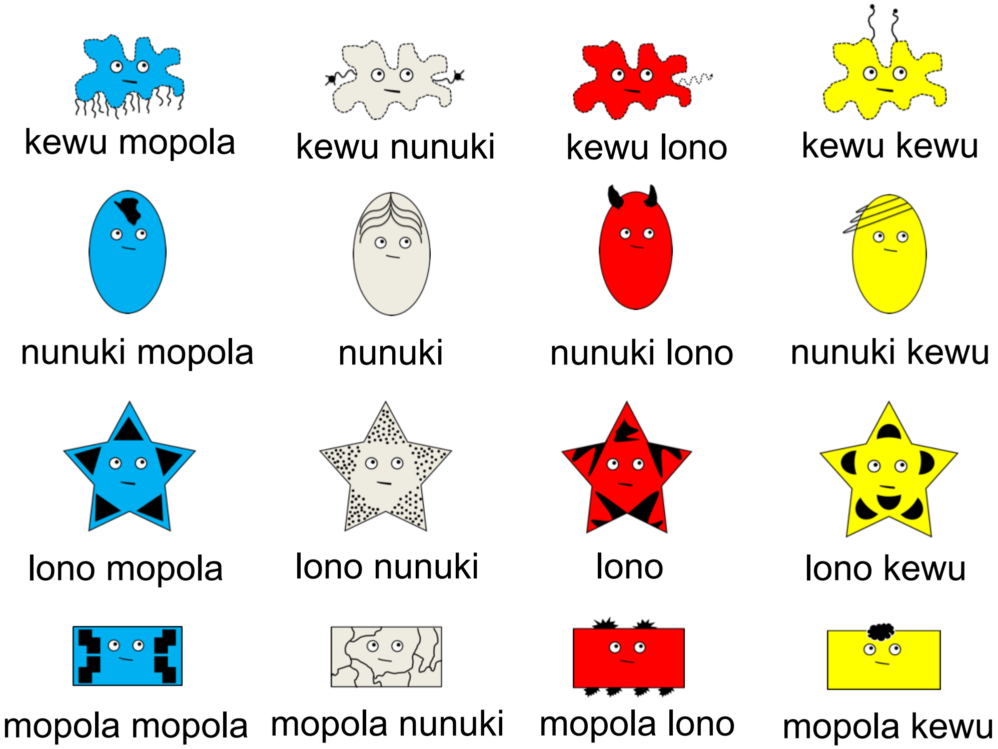

mege-ha
mege-hi
mege-hu
megi-lo
waka-ha
waka-hi
waka-hu
waki-lo
goko-ha
goko-hi
goko-hu
goki-lo
kuki-ha
kuko-hi
kuko-hu
kuki-lo

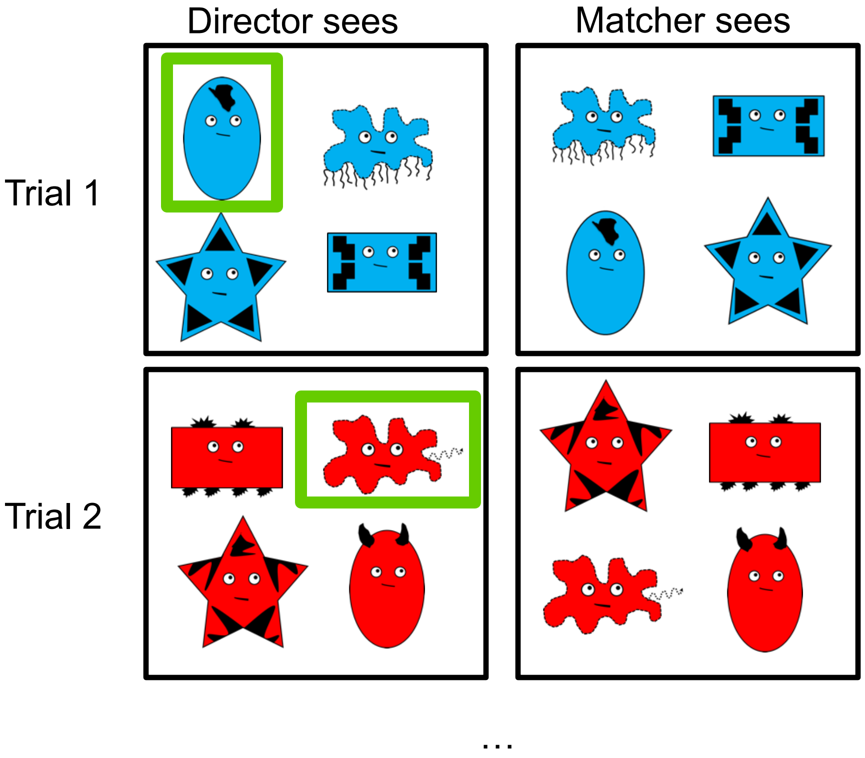

Director sees
Matcher sees
Trial 1
Trial 2
...

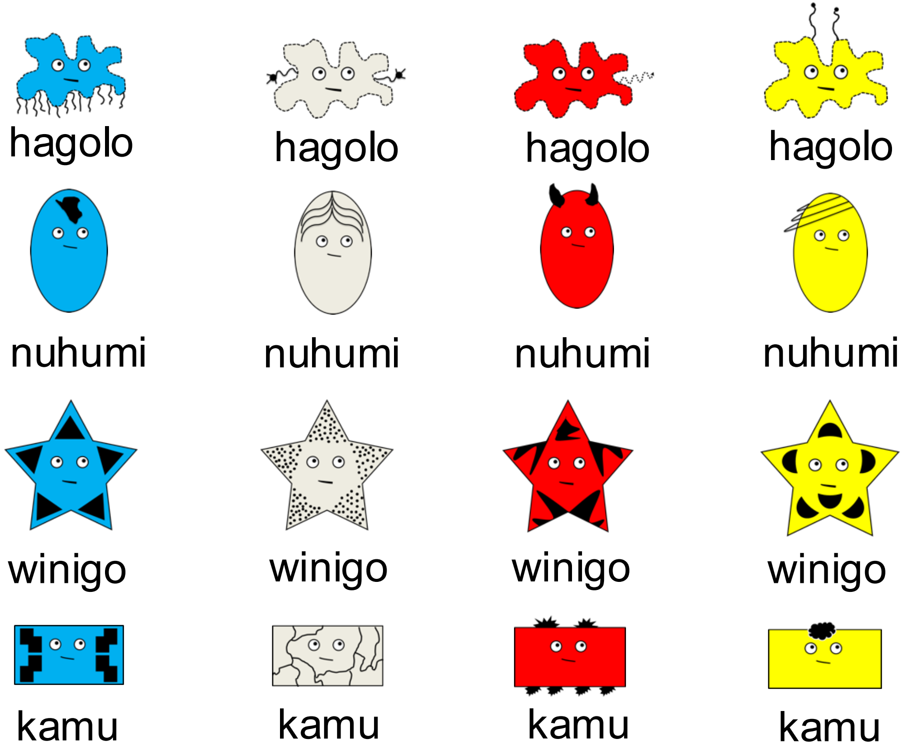
On shape-relevant trials …



On colour-relevant trials…

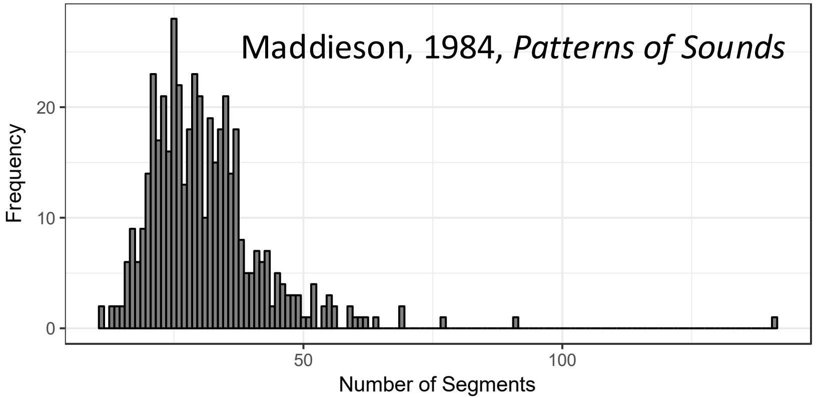



Director sees
Matcher sees
Trial 1
Trial 2
...

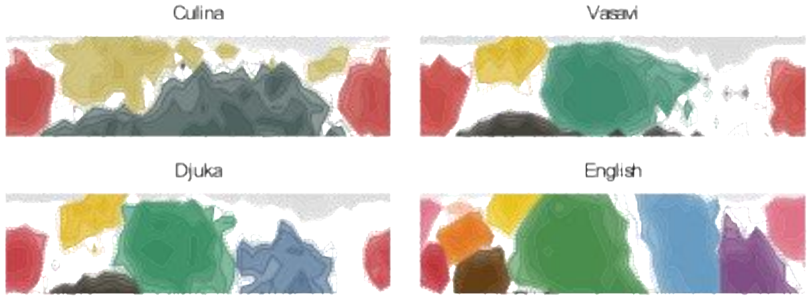

hagolo
hagolo
hagolo
hagolo
nuhumi
nuhumi
nuhumi
nuhumi
winigo
winigo
winigo
winigo
kamu
kamu
kamu
kamu

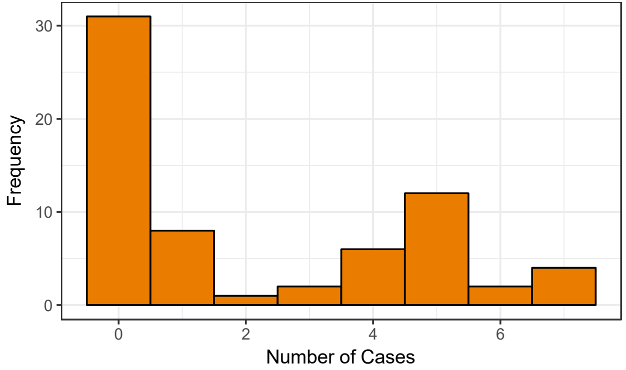
Adaptation to differing learning constraints

Languages differ in their complexity

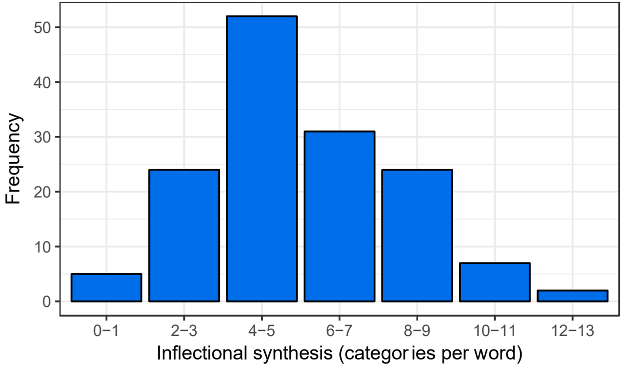


Zaslavsky et al, 2019, *Cognitive Neuropsychology*



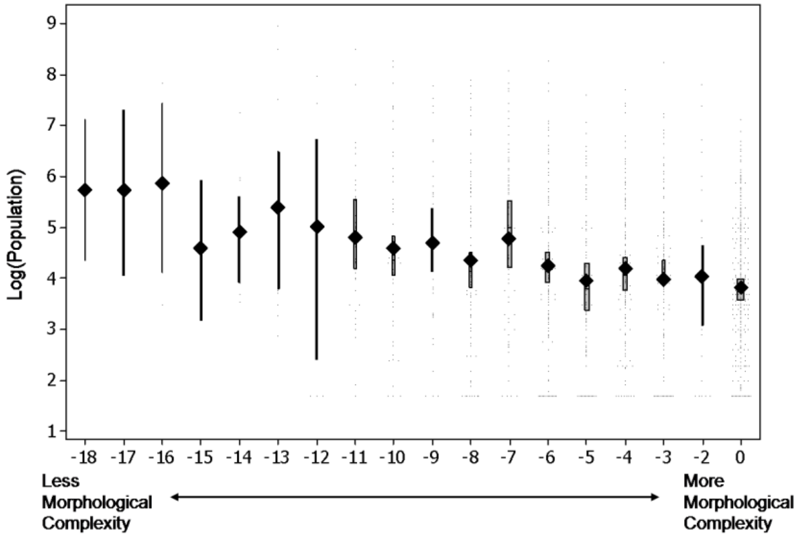


Bentz & Winter, 2013, *Language Dynamics & Change*

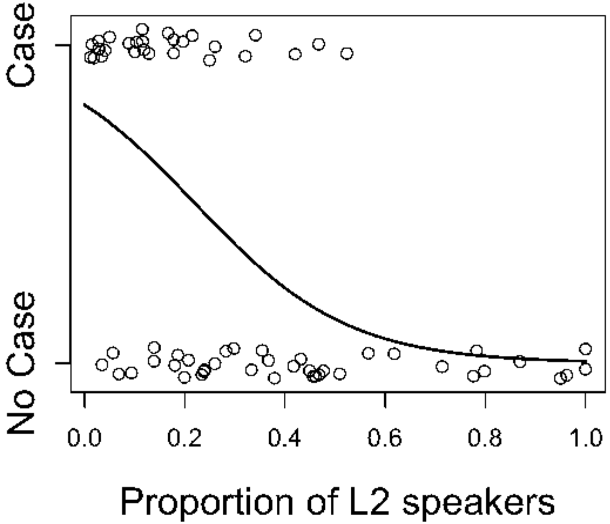


Bickel & Nichols, 2013, in *WALS*

Languages spoken in larger populations, with more non-native speakers, tend to be simpler*



Lupyan & Dale, 2010, *PLOS ONE*



Bentz & Winter, 2013, *LDC*

* Maybe: see also e.g. Koplenig, 2019, *Roy Soc Open Science;* Kauhanen, Einhaus & Walkden, 2023, *JoLE;* Koplenig, 2023, arXiv; Shcherbakova et al., 2023, *Science Advances*

Potential mechanism: uncertainty in interaction

e.g. Atkinson et al., 2017, *JoLE*; Raviv et al., 2019, *Proc Roy Soc*; Feher et al., 2019, *JML*; Frank & Smith, 2020, *Proc Cog Sci*; Loy et al., 2020, *Discourse Processes; …*

Potential mechanism:
learning from heterogenous sources;
learning from non-native speakers

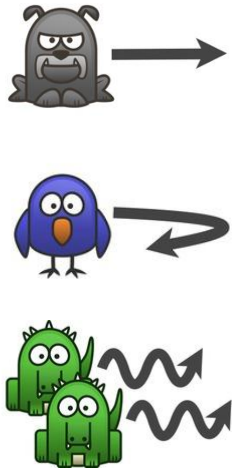
Atkinson et al., 2018, *Cognitive Science;* Raviv et al., 2019, *Proc Roy Soc;*
Berdicevskis & Semenuks, 2022, *PLOS ONE*; Smith, 2024, *Proc Cog Sci*

Two-stage experiment

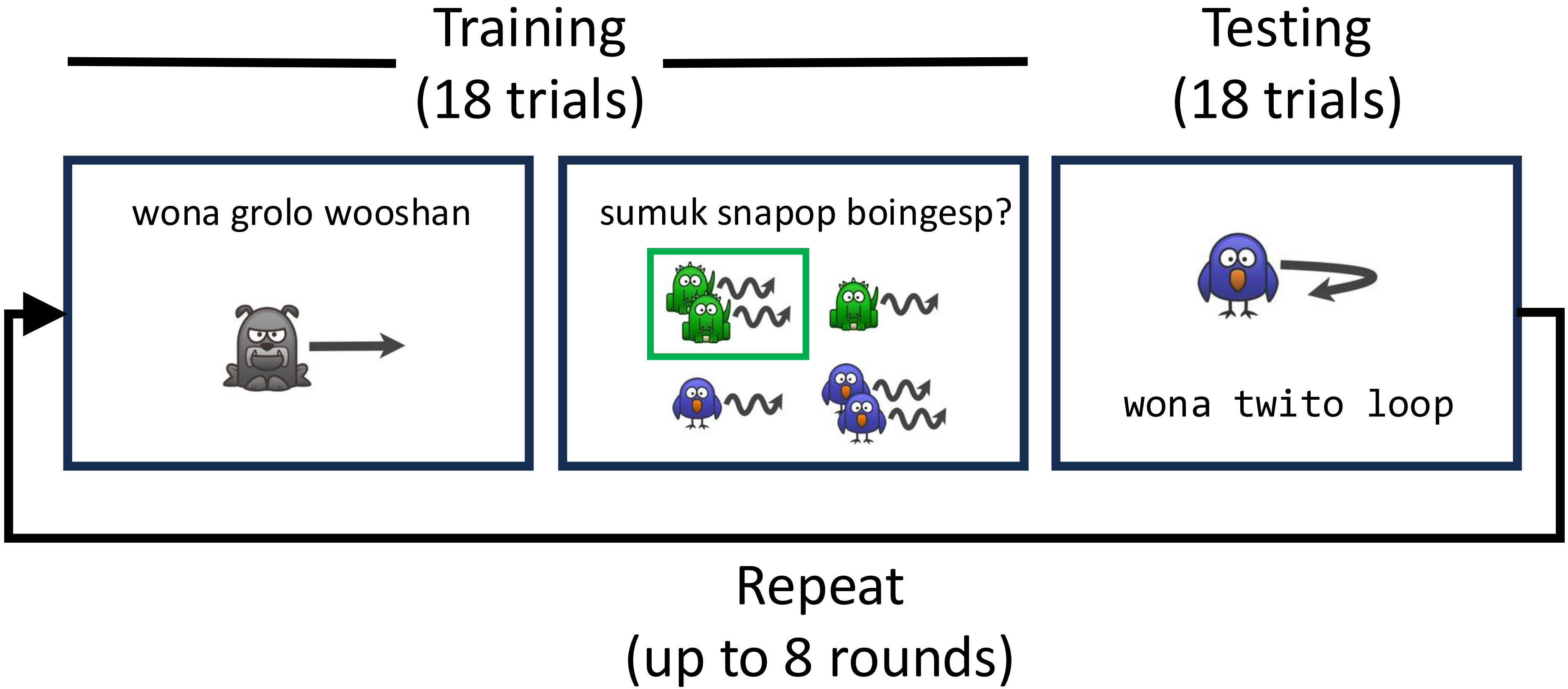
- Experiment 1: establishing an experimental proxy for native vs non-native speakers
- Experiment 2: investigating the effect of population size and proportion of non-native speakers on a single step of intergenerational transmission

Experiment 1: establishing an experimental proxy for native vs non-native speakers

Target language



Quantifier	Noun	Verb	
won-a	grol-o	woosh-an	"one dog straight"
sum-ak	grol-op	woosh-asp	"two dogs straight"
won-a	grol-o	boing-an	"one dog bounce"
sum-ak	grol-op	boing-asp	"two dogs bounce"
won-a	grol-o	loop-an	"one dog loop"
sum-ak	grol-op	loop-onk	"two dogs loop"
won-a	twit-o	woosh-an	"one bird straight"
sum-ak	twit-o	woosh-asp	"two birds straight"
won-a	twit-o	boing-an	"one bird bounce"
sum-ak	twit-o	boing-asp	"two birds bounce"
won-a	twit-o	loop-an	"one bird loop"
sum-ak	twit-o	loop-onk	"two birds loop"
won-u	snap-o	woosh-en	"one crocodile straight"
sum-uk	snap-op	woosh-esp	"two crocodiles straight"
won-u	snap-o	boing-en	"one crocodile bounce"
sum-uk	snap-op	boing-esp	"two crocodiles bounce"
won-u	snap-o	loop-en	"one crocodile loop"
sum-uk	snap-op	loop-onk	"two crocodiles loop"



Training
(18 trials)
Testing
(18 trials)
wona grolo wooshan
sumuk snapop boingesp?
wona twito loop
Repeat
(up to 8 rounds)

depends on number + animal

depends on number + animal

depends on number + animal + movement

Quantifier	Noun	Verb	
won-a	grol-o	woosh-an	"one dog straight"
sum-ak	grol-op	woosh-asp	"two dogs straight"
won-a	grol-o	boing-an	"one dog bounce"
sum-ak	grol-op	boing-asp	"two dogs bounce"
won-a	grol-o	loop-an	"one dog loop"
sum-ak	grol-op	loop-onk	"two dogs loop"
won-a	twit-o	woosh-an	"one bird straight"
sum-ak	twit-o	woosh-asp	"two birds straight"
won-a	twit-o	boing-an	"one bird bounce"
sum-ak	twit-o	boing-asp	"two birds bounce"
won-a	twit-o	loop-an	"one bird loop"
sum-ak	twit-o	loop-onk	"two birds loop"
won-u	snap-o	woosh-en	"one crocodile straight"
sum-uk	snap-op	woosh-esp	"two crocodiles straight"
won-u	snap-o	boing-en	"one crocodile bounce"
sum-uk	snap-op	boing-esp	"two crocodiles bounce"
won-u	snap-o	loop-en	"one crocodile loop"
sum-uk	snap-op	loop-onk	"two crocodiles loop"

depends on Ø

depends on number

depends on number + animal

Quantifier	Noun	Verb	
won-a	grol-o	woosh-an	"one dog straight"
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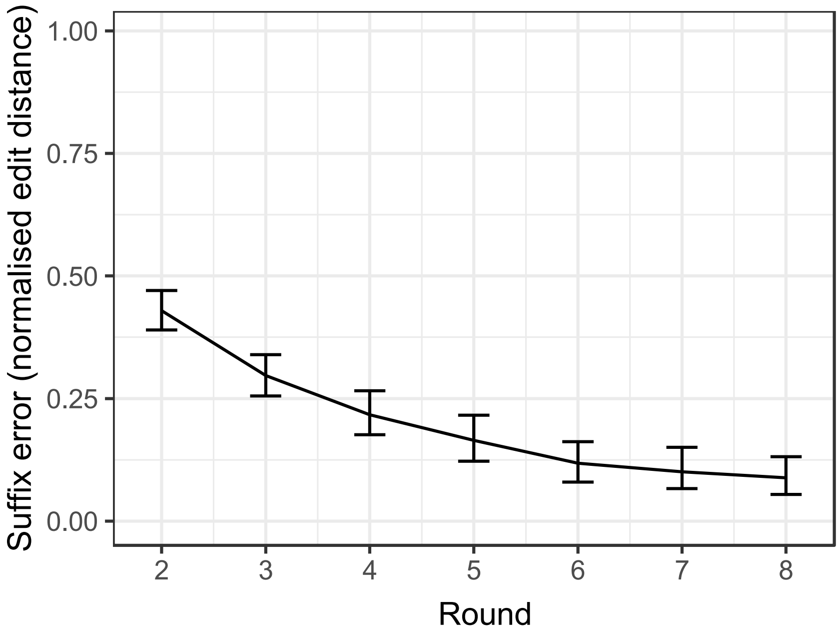
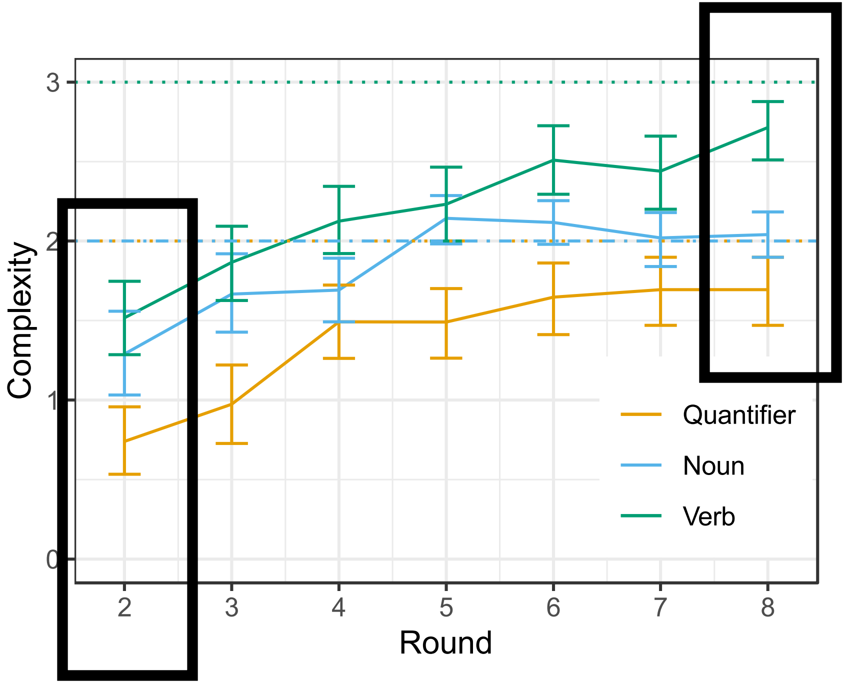
sum-a	snap-op	woosh-esp	"two crocodiles straight"
won-a	snap-o	boing-en	"one crocodile bounce"
sum-a	snap-op	boing-esp	"two crocodiles bounce"
won-a	snap-o	loop-en	"one crocodile loop"
sum-a	snap-op	loop-esp	"two crocodiles loop"

Experiment 1 results (N=94)



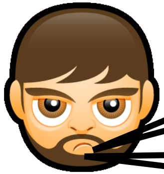
Proxy for native speakers (≈ no simplification)



Proxy for non-native speakers (simplification)

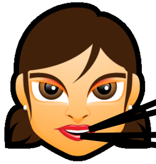
Experiment 2: investigating the effect of population size and proportion of non-native speakers on a single step of intergenerational transmission

Experiment 1 participants



Round 2
-a -o -an, -a -op -asp, -a -o -an, …

Round 8
-a -o -an, -ak -op -asp, -a -o -an, …



Round 2
-ak -o -an, -ak -op -esp, -a -o -esp, …

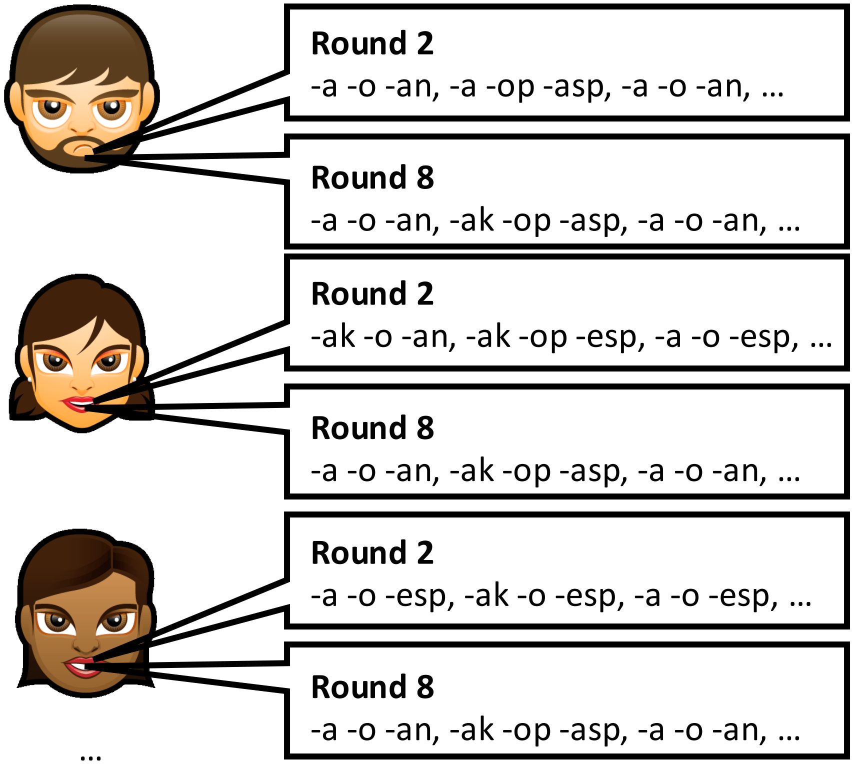
Round 8
-a -o -an, -ak -op -asp, -a -o -an, …

Round 2
-a -o -esp, -ak -o -esp, -a -o -esp, …

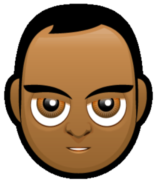
Round 8
-a -o -an, -ak -op -asp, -a -o -an, …

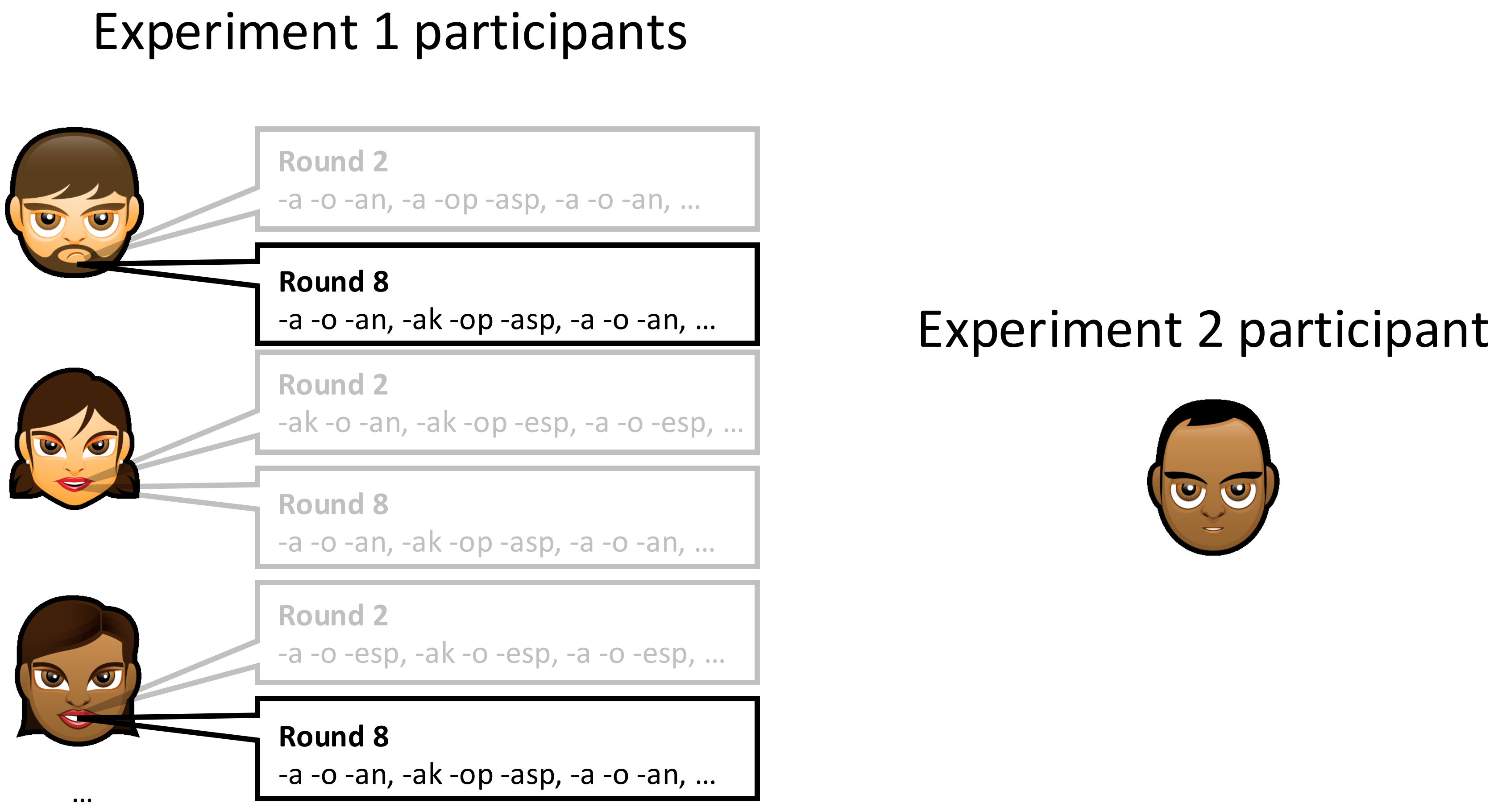
…

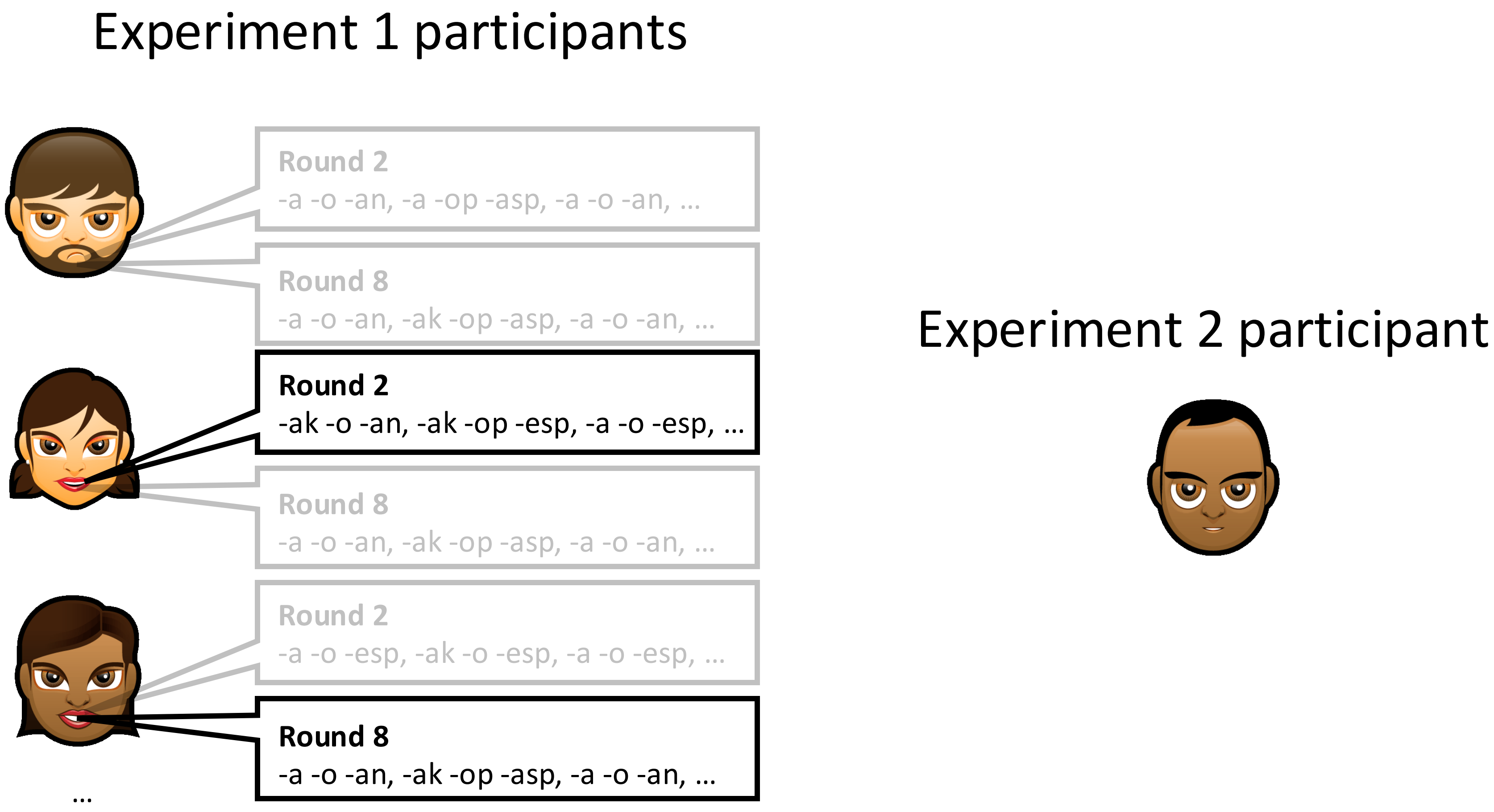
Experiment 1 participants

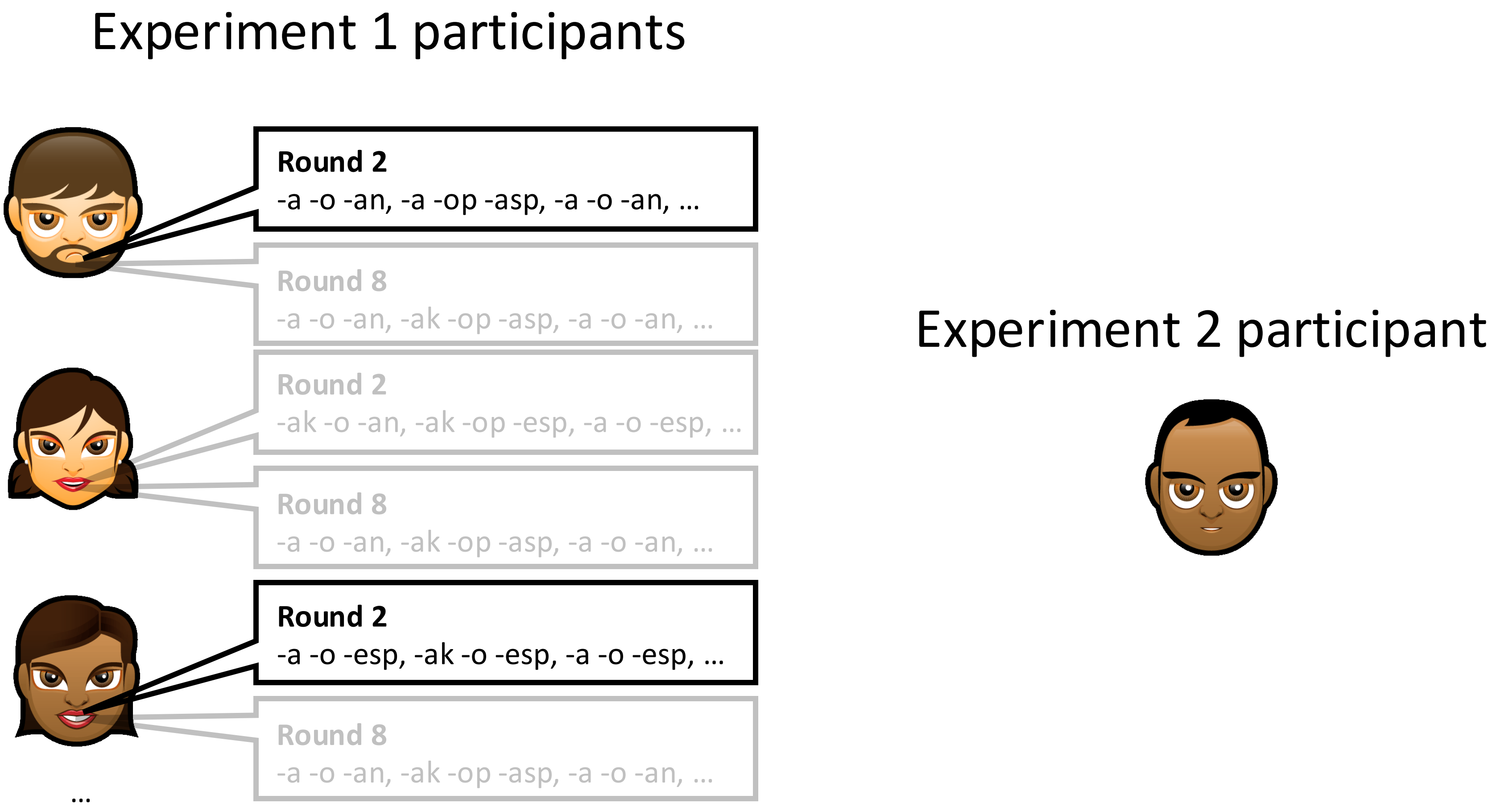


Experiment 2 participant




Experiment 1 participants
Round 2
-a -o -an, -a -op -asp, -a -o -an, …
Round 8
-a -o -an, -ak -op -asp, -a -o -an, …
Round 2
-ak -o -an, -ak -op -esp, -a -o -esp, …
Round 8
-a -o -an, -ak -op -asp, -a -o -an, …
Round 2
-a -o -esp, -ak -o -esp, -a -o -esp, …
Round 8
-a -o -an, -ak -op -asp, -a -o -an, …
…
Experiment 2 participant


Experiment 1 participants
Round 2
-a -o -an, -a -op -asp, -a -o -an, …
Round 8
-a -o -an, -ak -op -asp, -a -o -an, …
Round 2
-ak -o -an, -ak -op -esp, -a -o -esp, …
Round 8
-a -o -an, -ak -op -asp, -a -o -an, …
Round 2
-a -o -esp, -ak -o -esp, -a -o -esp, …
Round 8
-a -o -an, -ak -op -asp, -a -o -an, …
…
Experiment 2 participant


Experiment 1 participants
Round 2
-a -o -an, -a -op -asp, -a -o -an, …
Round 8
-a -o -an, -ak -op -asp, -a -o -an, …
Round 2
-ak -o -an, -ak -op -esp, -a -o -esp, …
Round 8
-a -o -an, -ak -op -asp, -a -o -an, …
Round 2
-a -o -esp, -ak -o -esp, -a -o -esp, …
Round 8
-a -o -an, -ak -op -asp, -a -o -an, …
…
Experiment 2 participant

Experiment 2 (N=522)

Small (2) or Large (8) population

All Native, Mixed or All Non-native input

	All Native	Mixed	All Non-native
Small	Labels from 2 Round 8 learners	Labels from 1 Round 2 and 1 Round 8 learner	Labels from 2 Round 2 learners
Large	Labels from 8 Round 8 learners	Labels from 4 Round 2 and 4 Round 8 learners	Labels from 8 Round 2 learners

Experiment 2 (N=522)

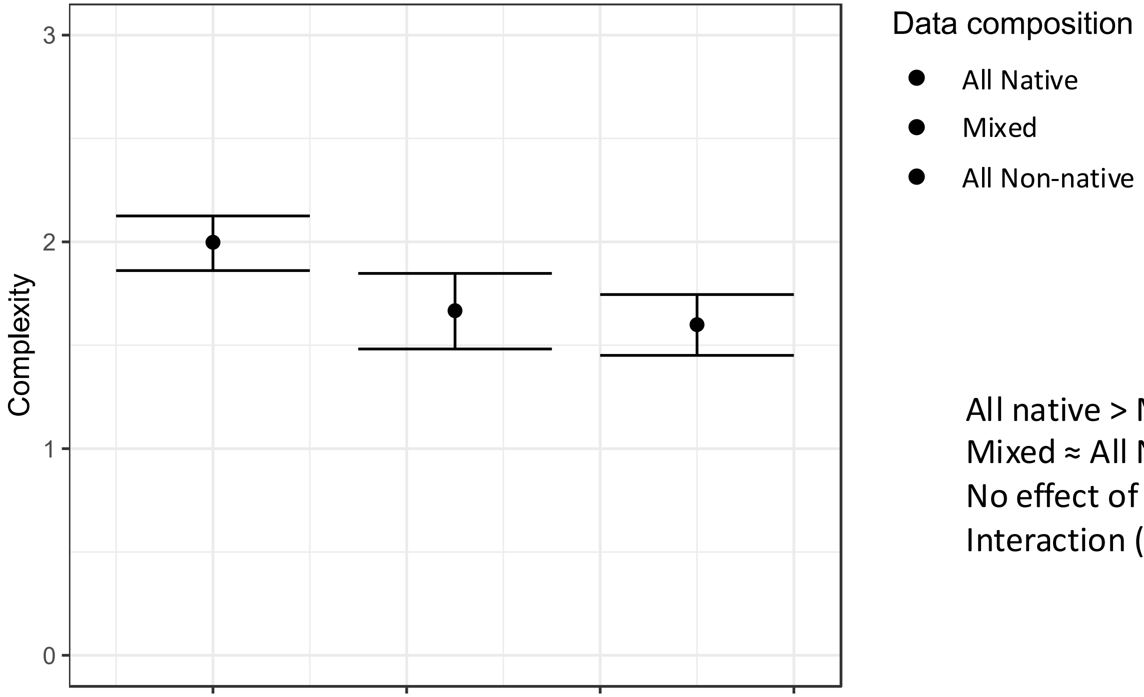
Small (2) or Large (8) population

All Native, Mixed or All Non-native input

	All Native	Mixed	All Non-native
Small	Labels from 2 Round 8 learners	Labels from 1 Round 2 and 1 Round 8 learner	Labels from 2 Round 2 learners
Large	Labels from 8 Round 8 learners	Labels from 4 Round 2 and 4 Round 8 learners	Labels from 8 Round 2 learners

Atkinson et al., 2018, *Cognitive Science:* no effects of input composition or population size (mixing effects? under-powered?)

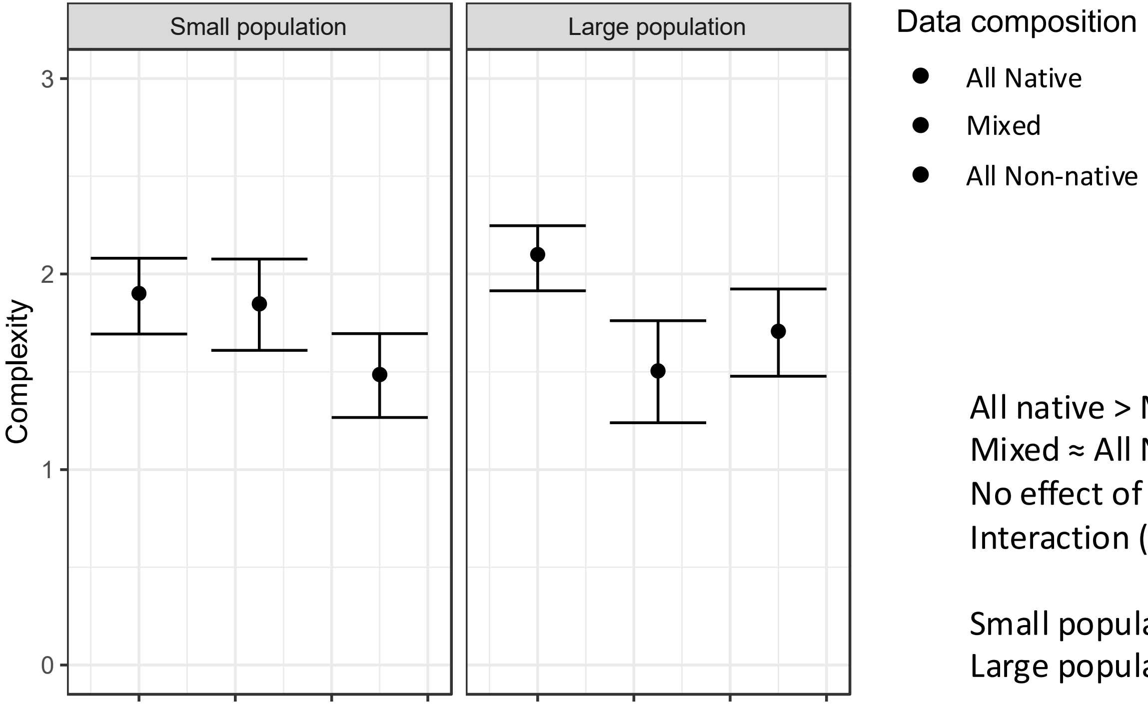
Experiment 2 results



All native > Mixed (b=-0.63, SE=0.22, p=.004)
Mixed ≈ All Non-native (b=-0.18, SE=0.22, p=.43)
No effect of population size (b=0.16, SE=0.18, p=.38)
Interaction (b=0.96, SE=0.45, p=.031)

Plot shows only round 8 data (N=222)
Stats use all N=522

Experiment 2 results



Plot shows only round 8 data (N=222)
Stats use all N=522

All native > Mixed (b=-0.63, SE=0.22, p=.004)
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No effect of population size (b=0.16, SE=0.18, p=.38)
Interaction (b=0.96, SE=0.45, p=.031)

Small population: Mixed-All Non-native contrast is significant
Large population: All Native-Mixed contrast is significant

The core of the argument (from Pinker & Bloom, 1990)

"All we have argued is that human language, like other specialized ***biological*** *systems, evolved by natural selection. Our conclusion is based on two facts that we would think would be entirely uncontroversial: Language shows signs of complex design for the communication of propositional structures, and the* ***only explanation*** *for the origin of organs with complex design is the process of natural selection."* (p. 726)

Pinker, S., & Bloom, P. (1990). Natural language and natural selection. *Behavioural and Brain Sciences, 13,* 707-784.

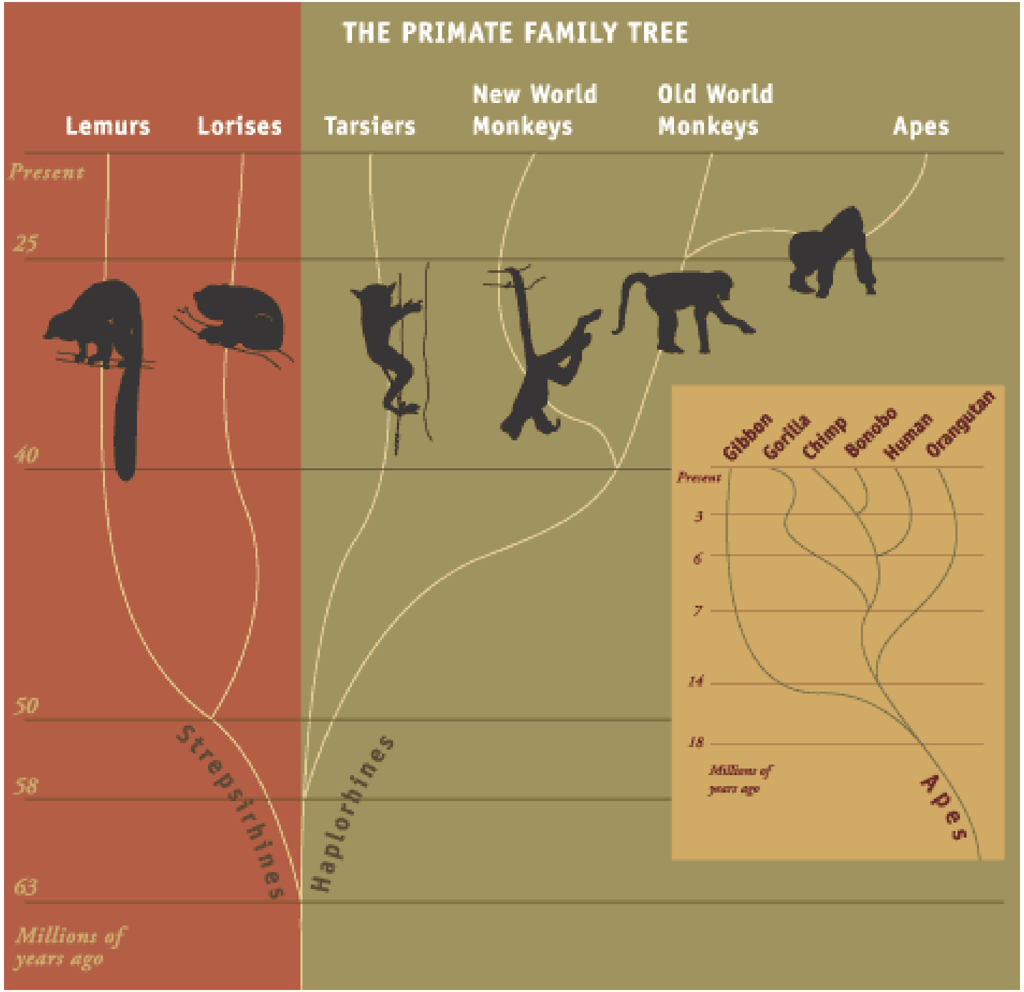
On the contrary…

One you have a communication system being transmitted by learning and use, structure develops “for free”

- Through the same processes we know drive language change (efficient communication, “system pressure” from analogy)
- Leading to adaptation, the appearance of design

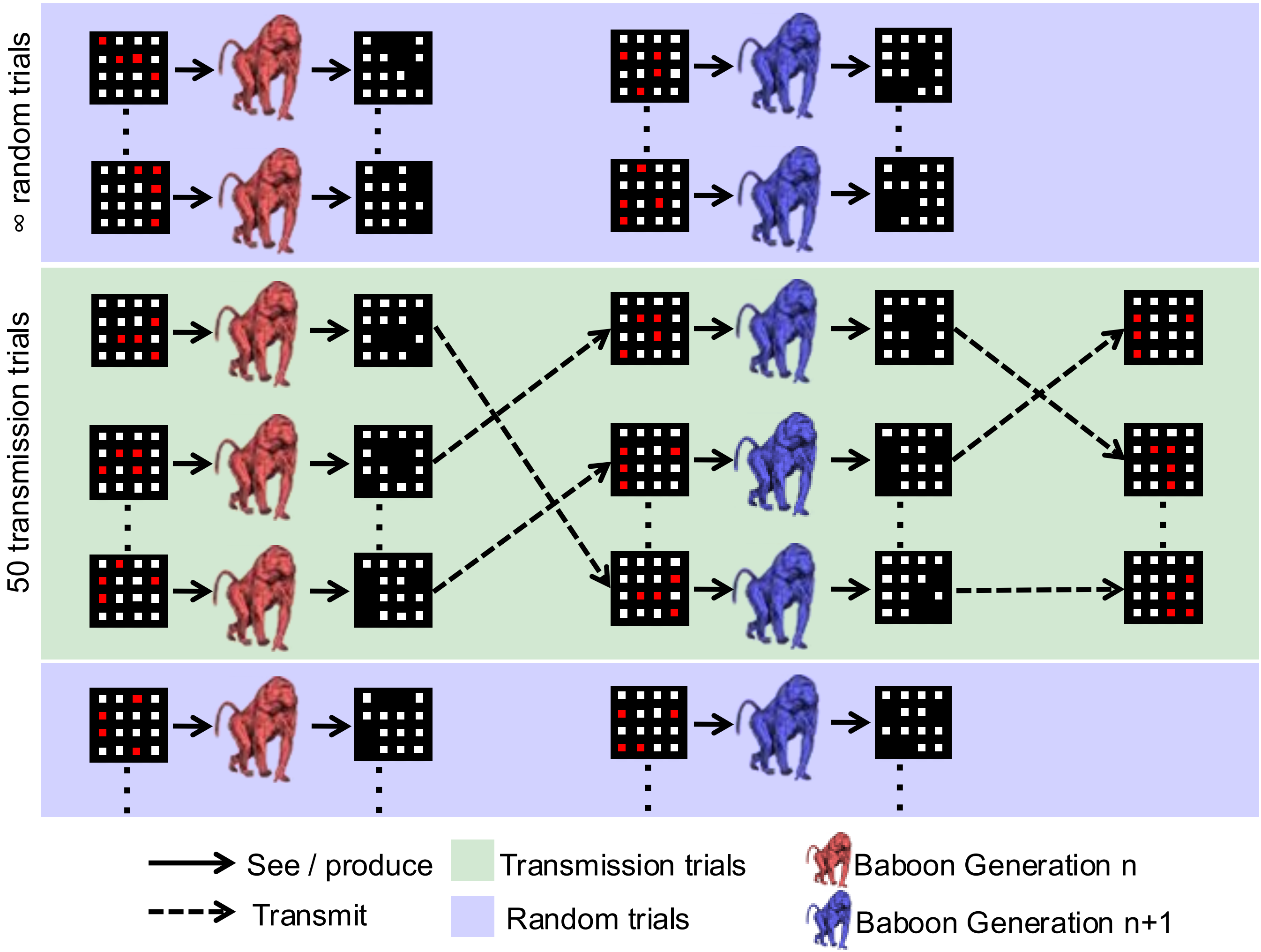
This changes the evolutionary question: we should ask, what cognitive prerequisites are required for iterated learning, and how did **those** evolve?

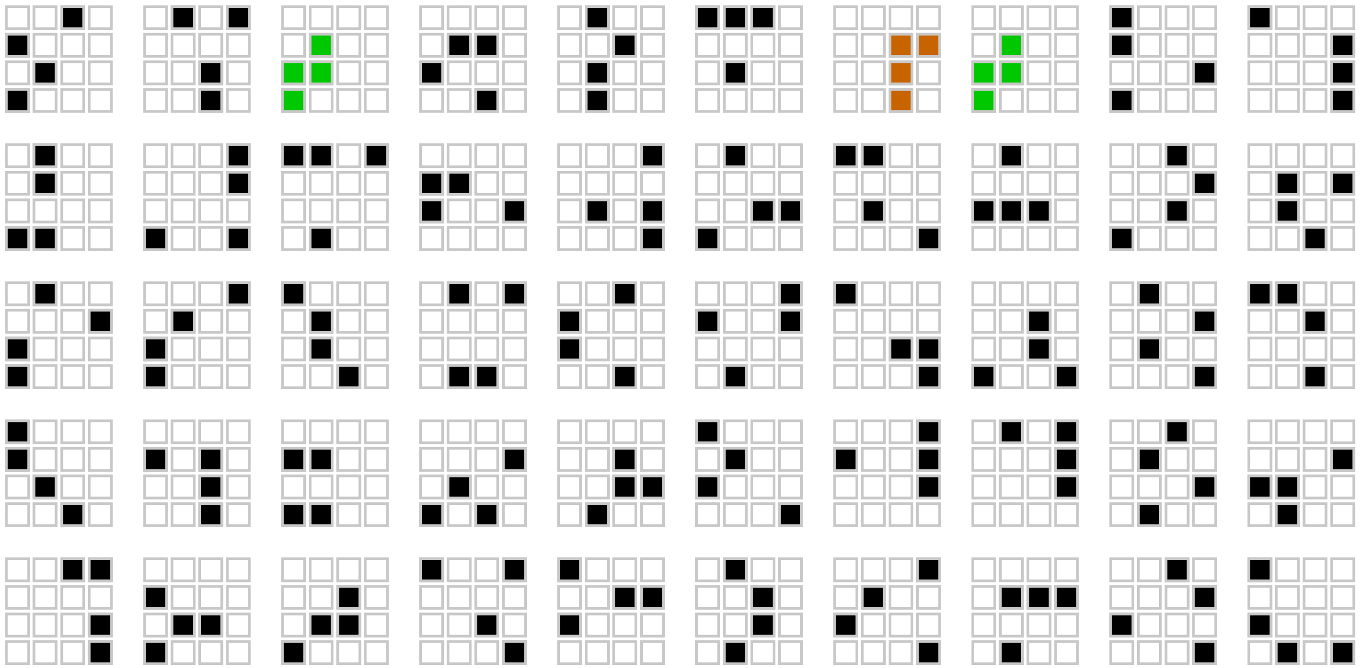
Would we see the same sort of processes unfolding if we ran iterated learning in non-human primates?



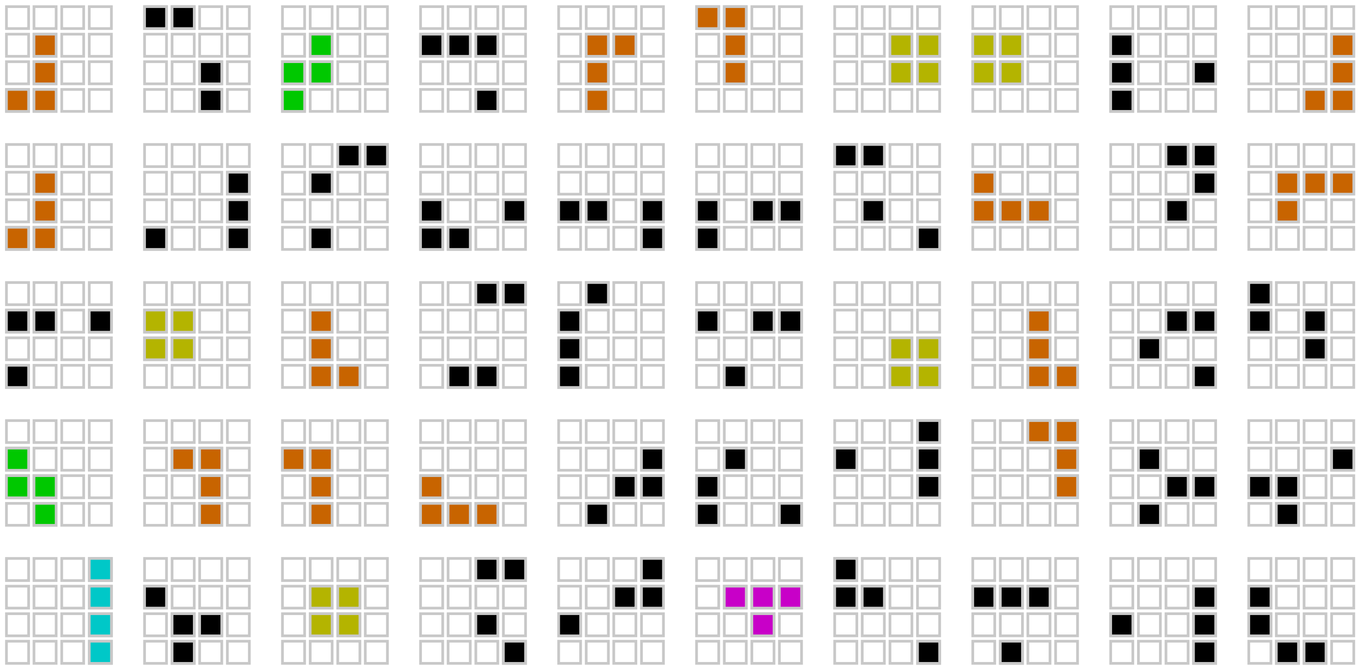
http://www.daviddarling.info/images/primate_family_tree.gif

Claidière, N., Smith, K., Kirby, S., & Fagot, J. (2014). Cultural evolution of a systematically structured behaviour in a non-human primate. *Proceedings of the Royal Society B, 281,* 20141541.

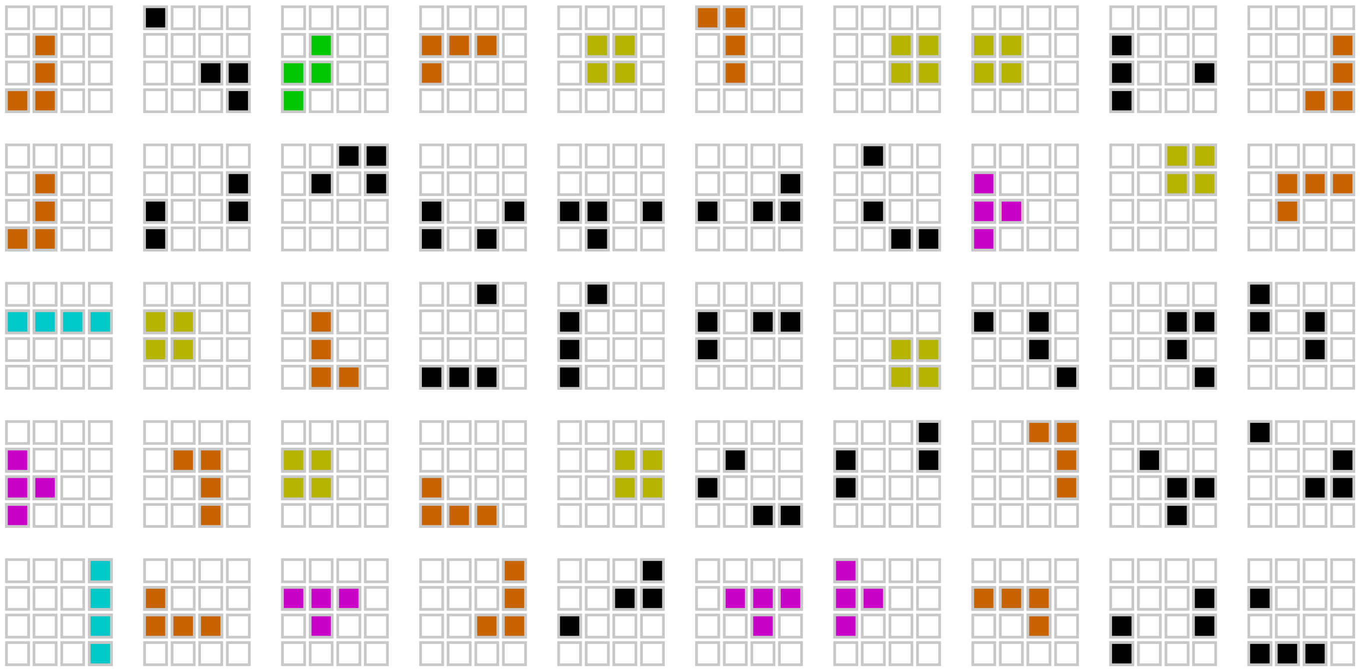

∞ random trials
50 transmission trials
See / produce
Transmit
Transmission trials
Random trials
Baboon Generation n
Baboon Generation n+1



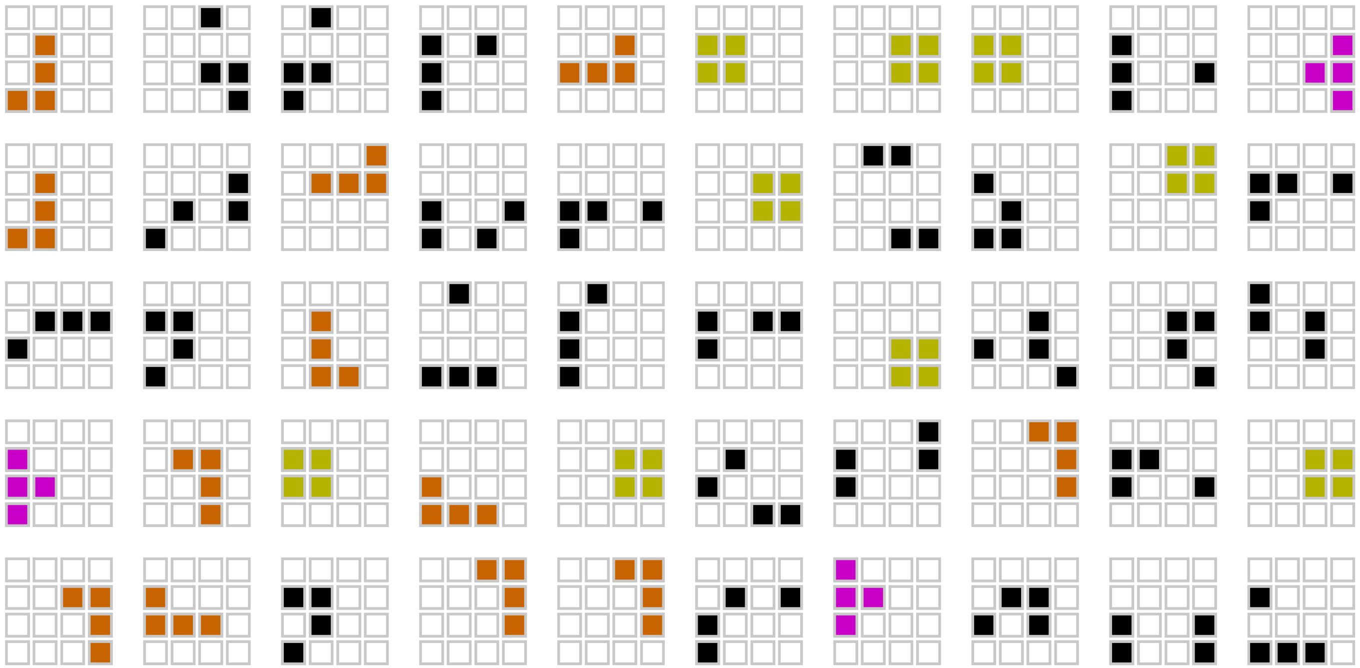
Random grids



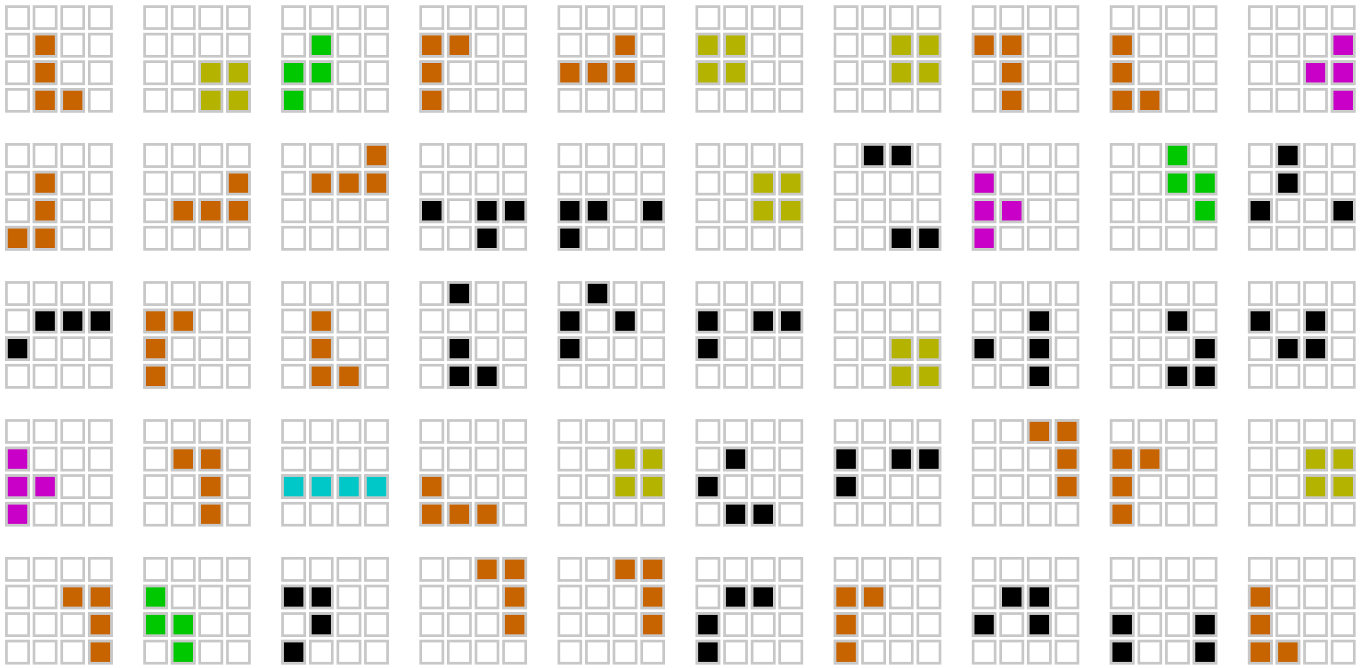
Generation 1



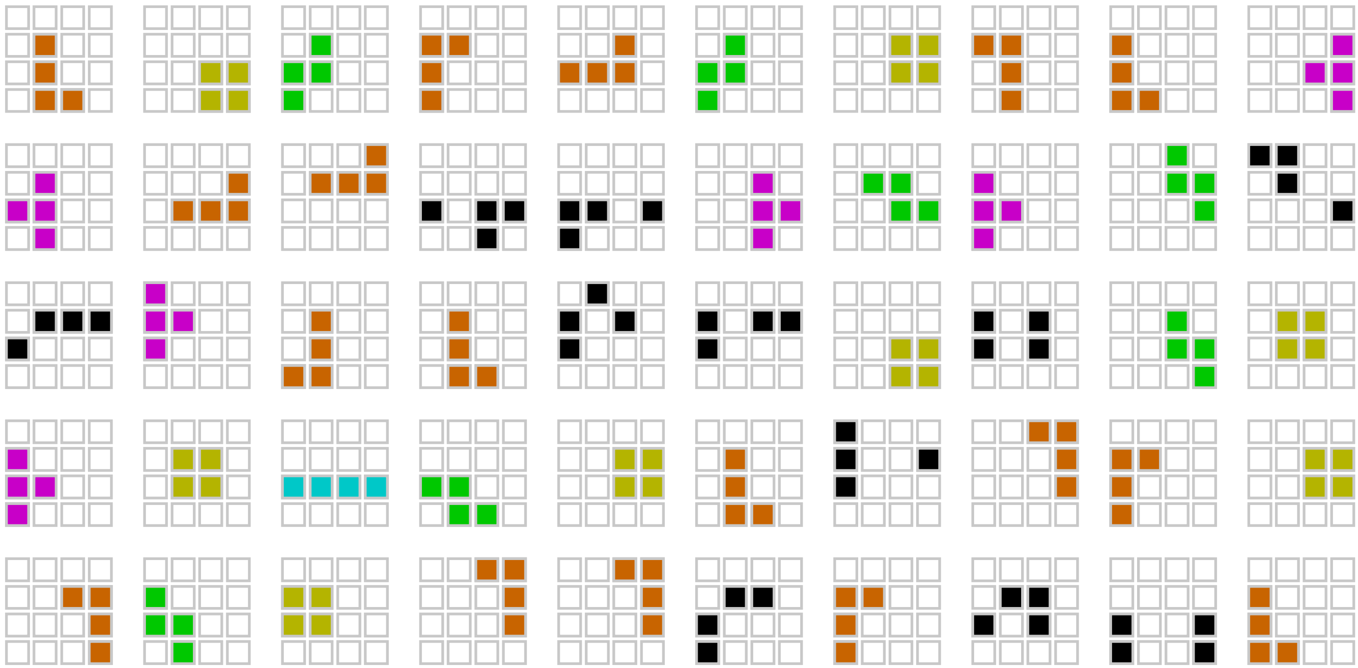
Generation 2



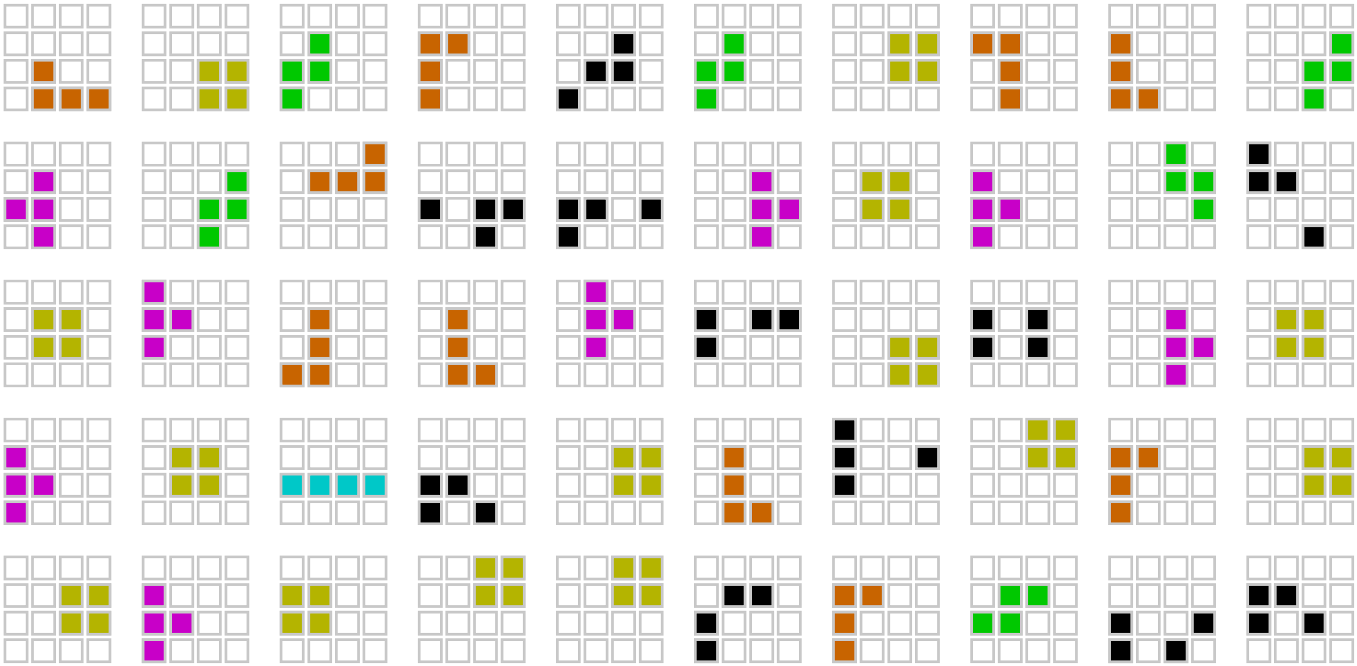
Generation 3



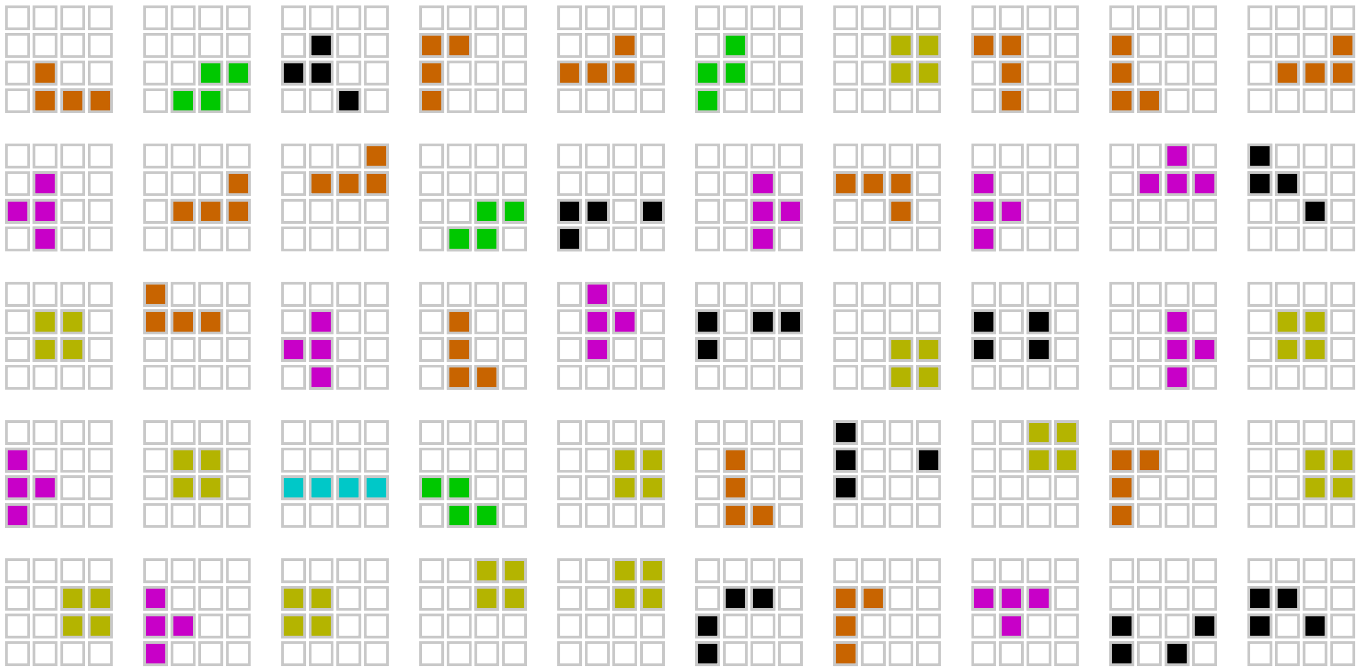
Generation 4



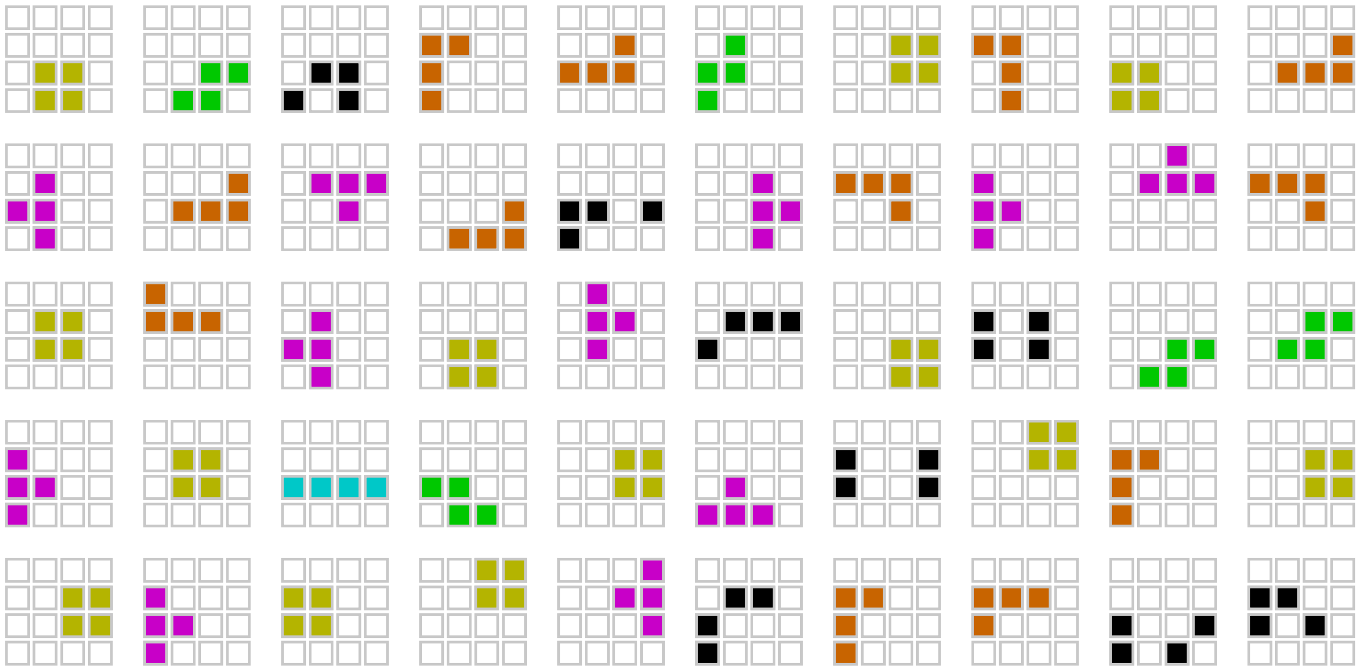
Generation 5



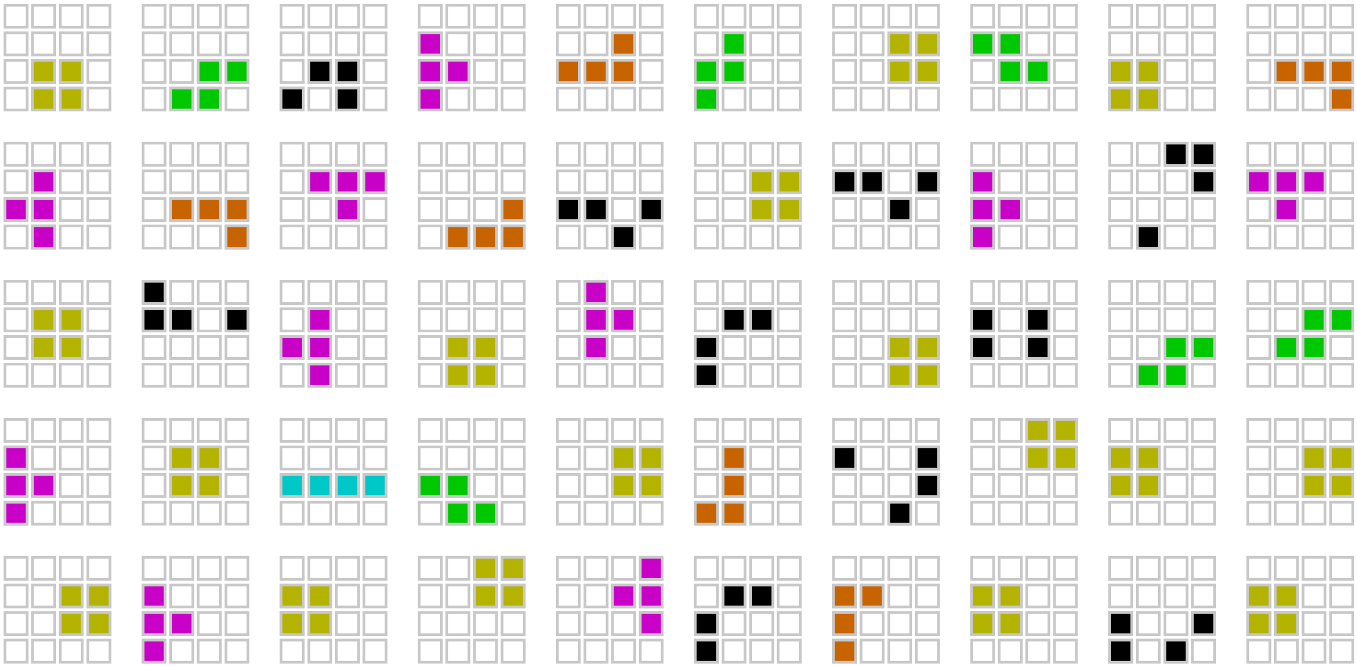
Generation 6



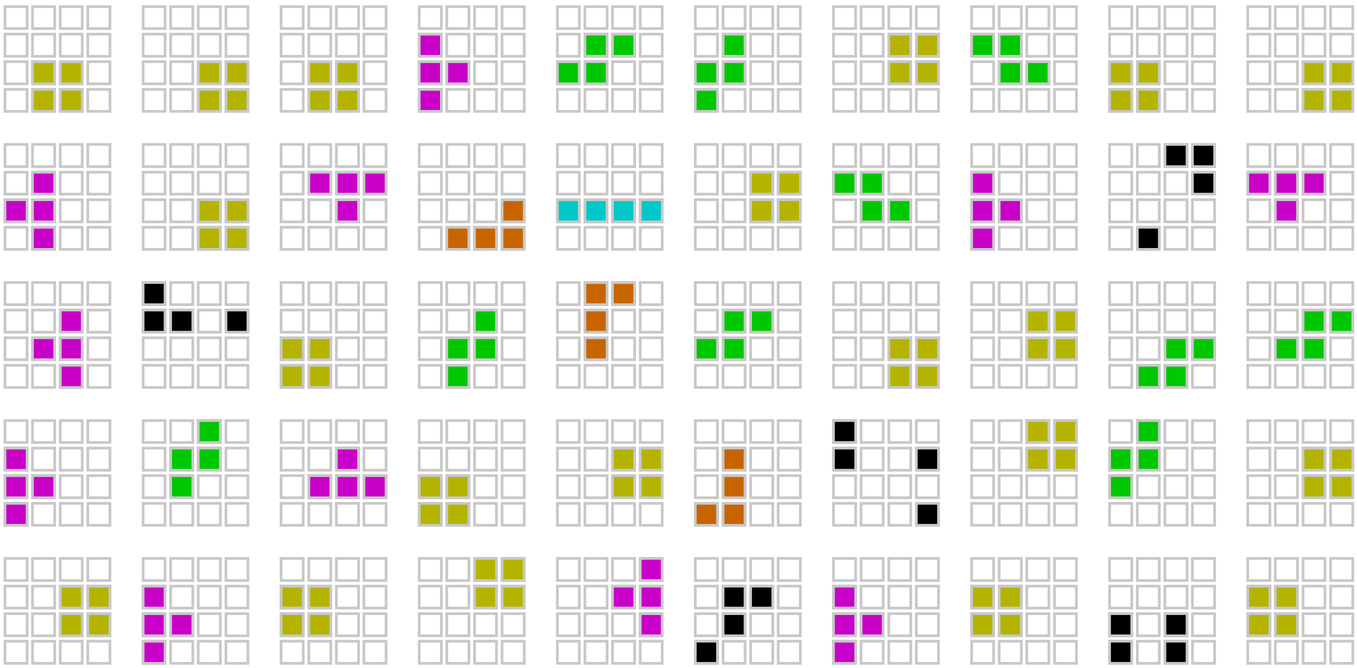
Generation 7



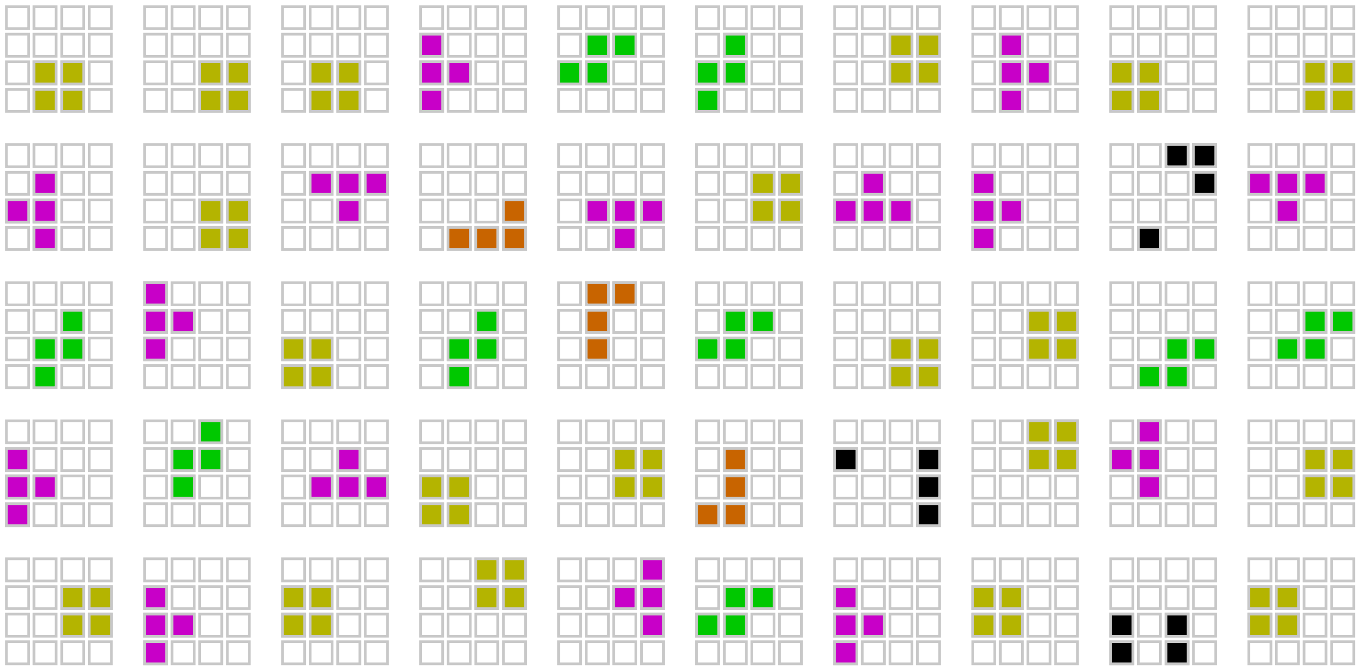
Generation 8



Generation 9



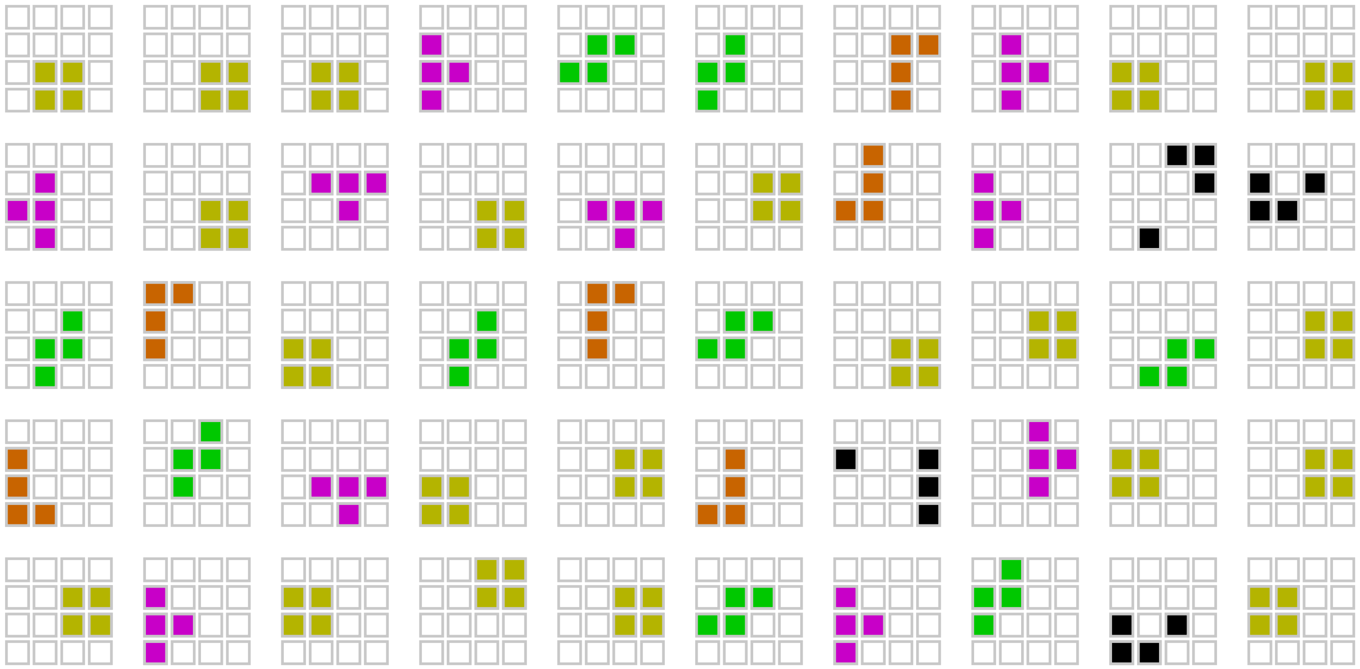
Generation 10



Generation 11

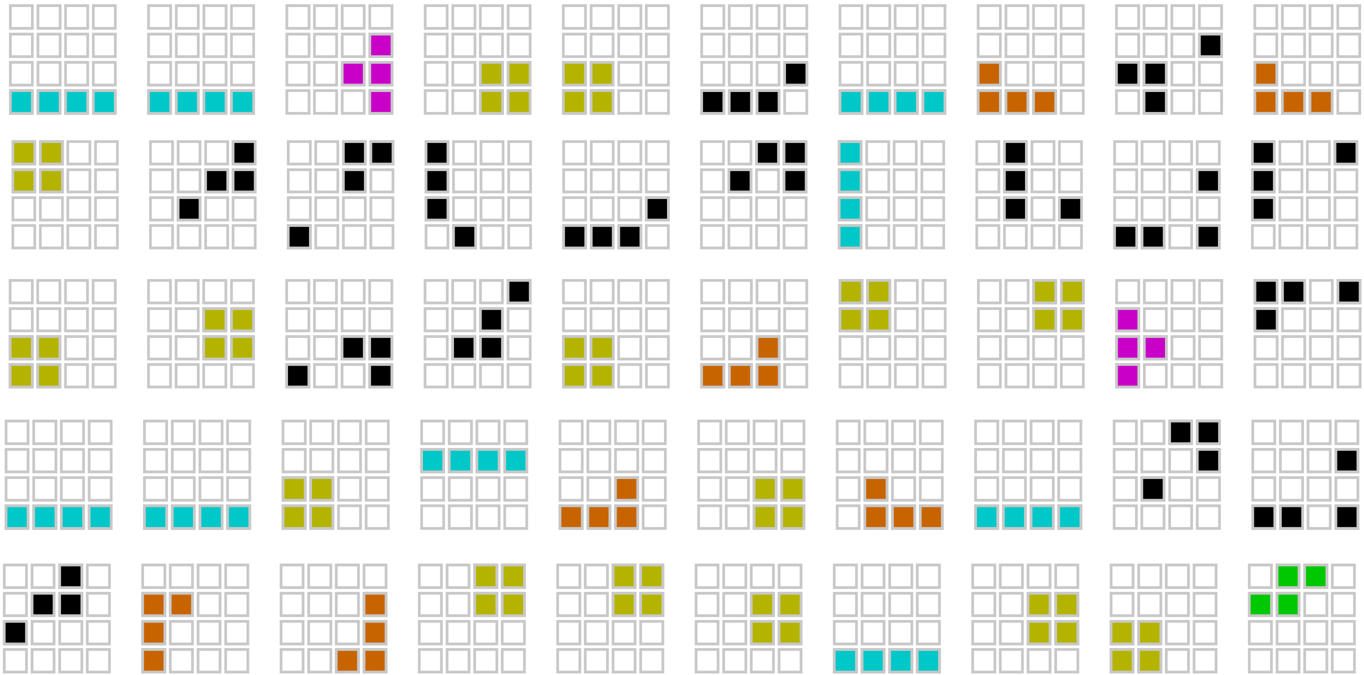
Generation 12

Emergence of a **system**



Chain 4, Generation 12

Emergence of a **system**



Chain 1, Generation 12

Systematic structure develops even in baboons (if you scaffold their environment in the right way)

Next class: cognitive prerequisites

- What kind of mind do you need to sustain a communication system by iterated learning?
- What can we learn about the evolution of those capacities in humans by studying the minds of other animals?